

Epigenetics assays

Analysis of gene expression regulation

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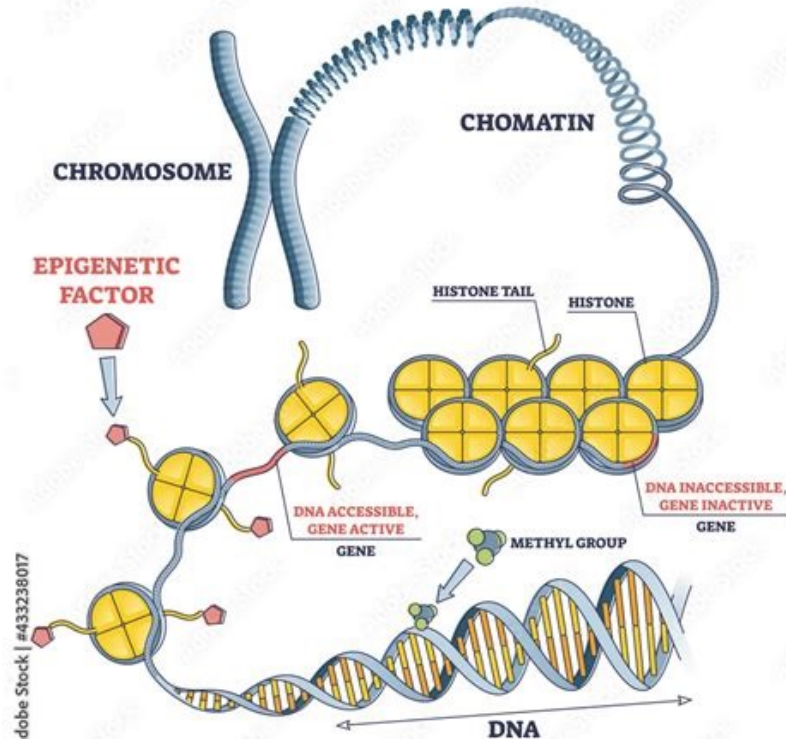


Epigenetics

- Epigenetics is the study of changes in gene function that are mitotically and/or meiotically heritable and that do not entail a change in DNA sequence. (Wu et al, 2001)
- Epigenetic mechanisms such as DNA methylation, Histone Post-Translational Modification (PTM), Nucleosome positioning are affected by development (in utero, childhood), environmental chemicals, drugs/pharmaceuticals, aging, diet,...

Technologies for epigenetics analysis

EPIGENETIC MECHANISMS

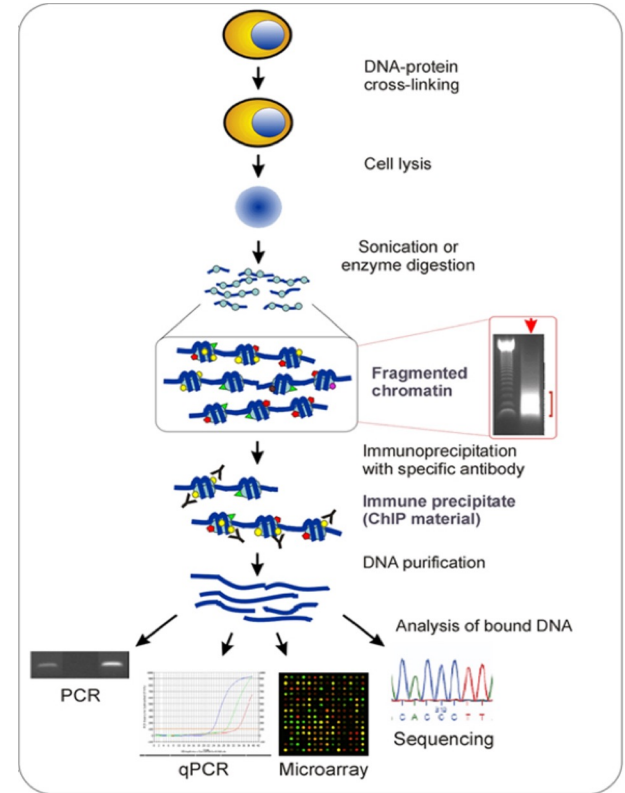


How to locate protein positions in the genome

Locate protein positions in the genome

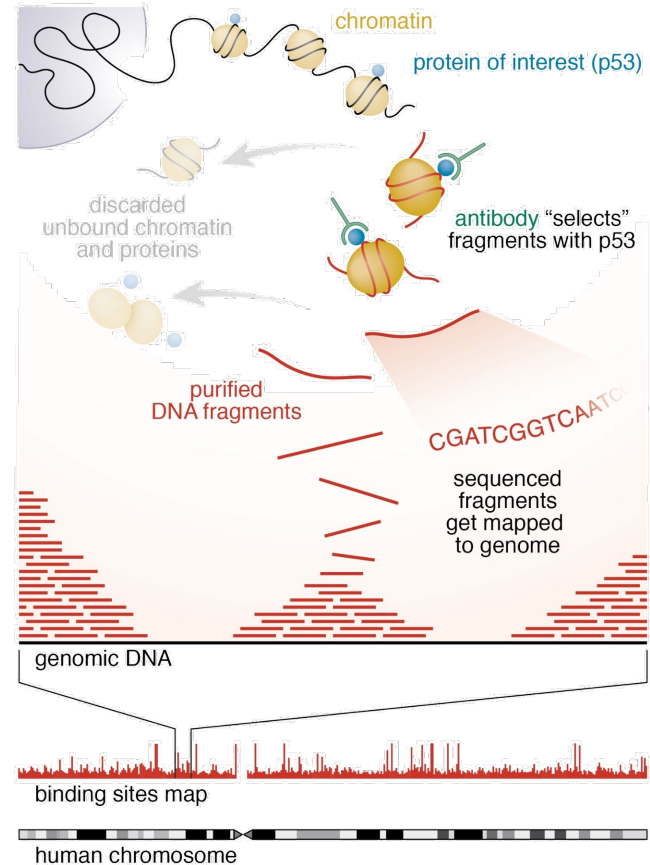
Differences in **methods to detect the bound DNA**

- small-scale: PCR / qPCR
- microarray: ~~ChIP-on-chip~~
- sequencing: **ChIP-seq, Cut&Tag, Cut&Run**

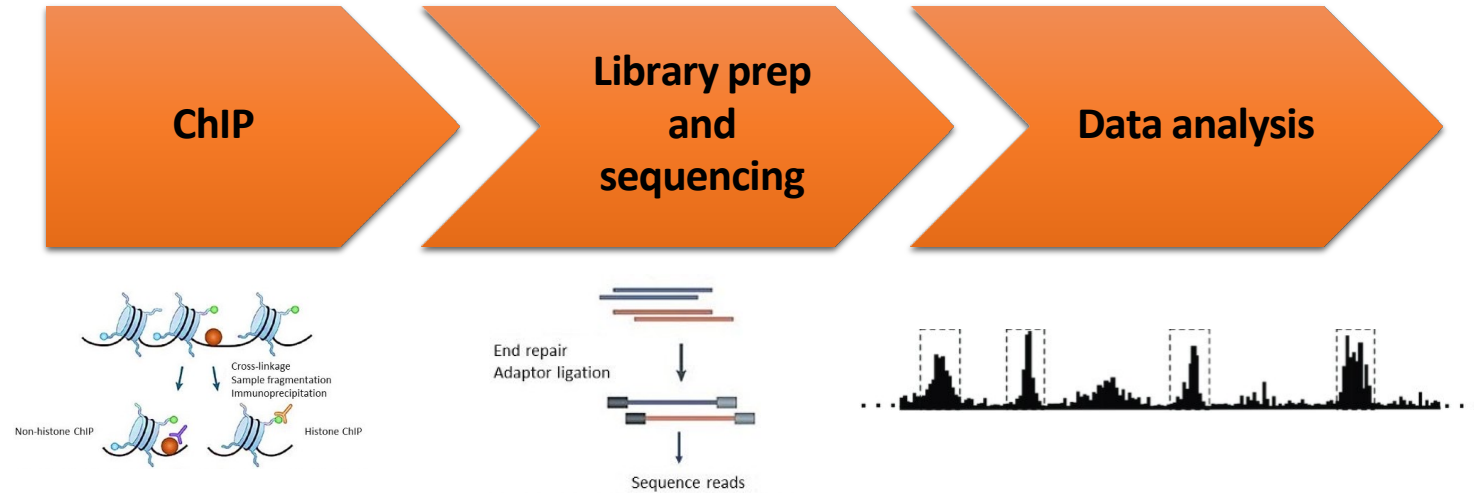


ChIP-seq

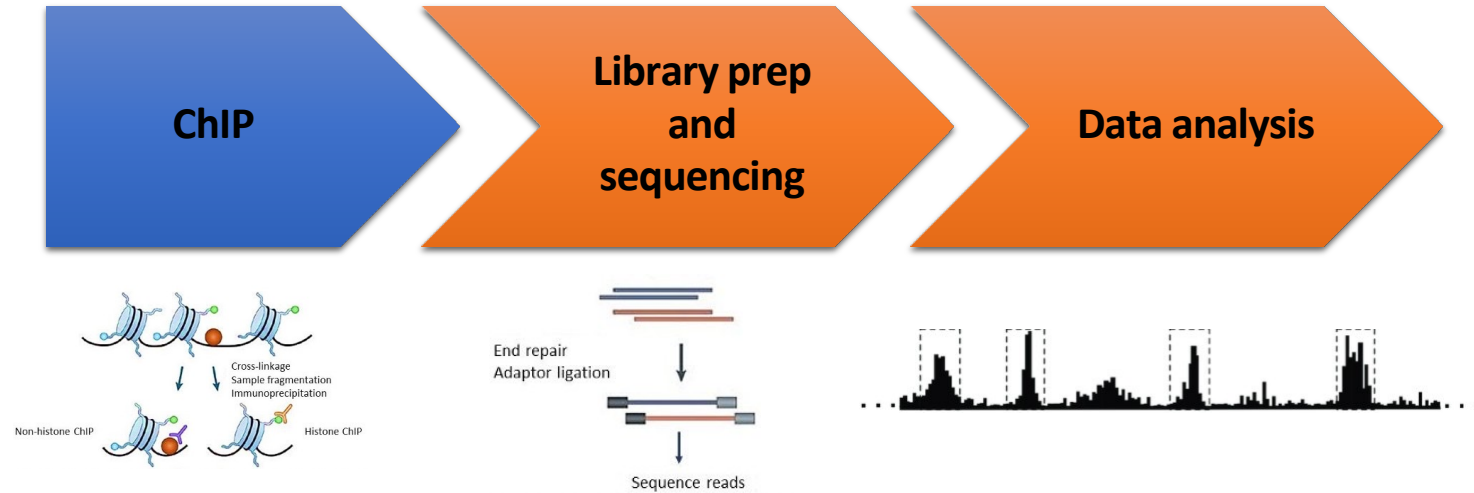
- Johnson et al, 2007
- Alternative to ChIP-on-chip (hybridization technique)
- ChIP-seq combines chromatin immunoprecipitation and sequencing to analyze protein interactions with DNA
- It can be used to detect and analyze
 - Binding sites of various proteins bound to DNA such as transcription factors (TF ChIP-seq)
 - Position of histone post translational modifications (Histone ChIP-seq)
 - Nucleotide modification such methylation (MeDIP-seq)
- Expected results of a ChIP-seq experiment are genomic regions with significant sequenced read enrichments (also called peaks)



ChIP-seq process

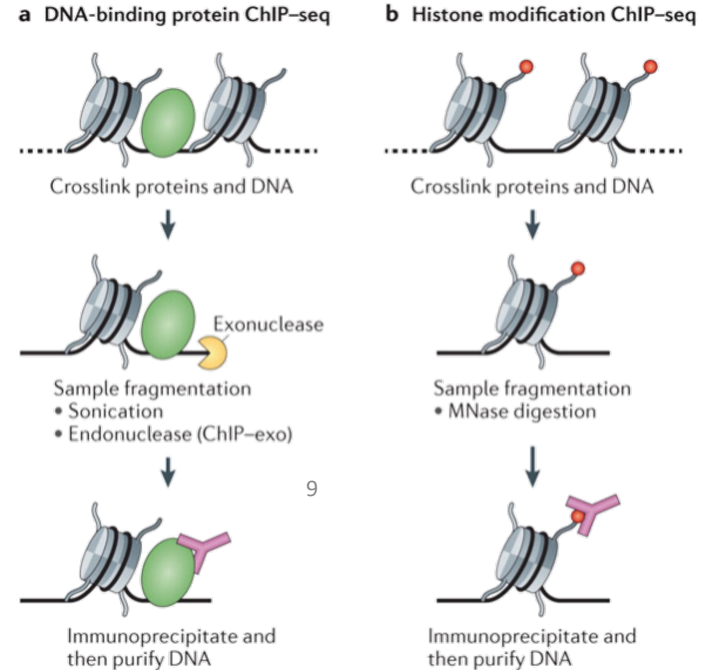


Chromatin ImmunoPrecipitation

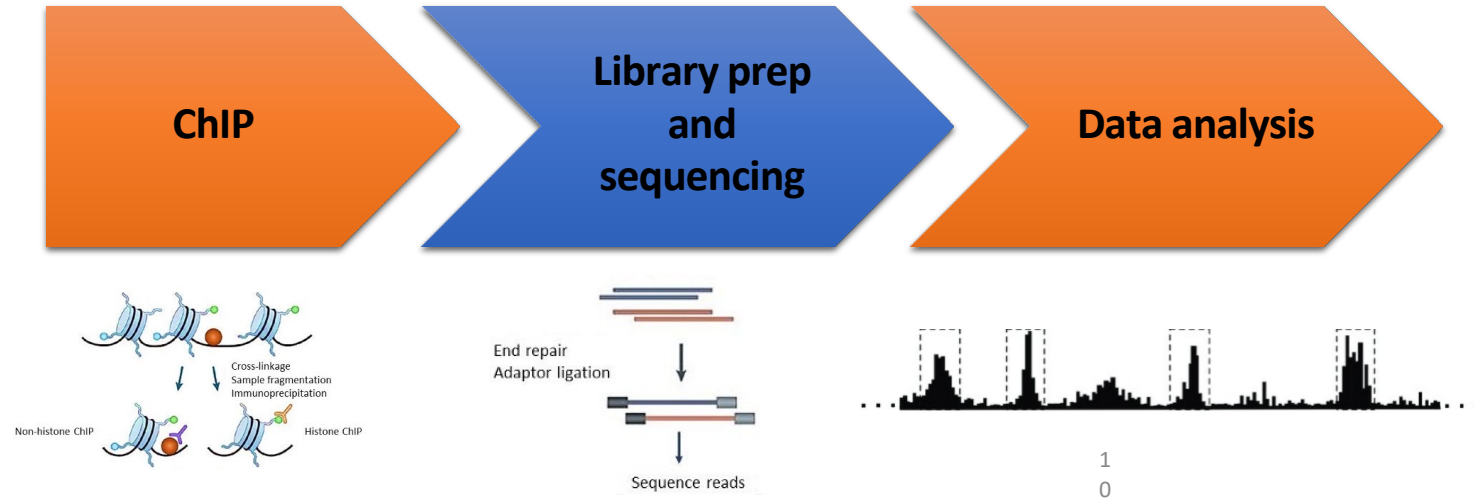


Chromatin ImmunoPrecipitation

- ChIP is used to isolate DNA fragments bound by proteins or modified histones of interest using an antibody specifically design to target them
- It requires a high number of cells (1-10 million cells)
- It requires highly tested and verified antibody working specifically for ChIP



Library prep and sequencing





Experimental design

ENCODE

- The Encyclopedia of DNA Elements (ENCODE) Consortium has carried out thousands of ChIP-seq experiments and has used this experience to develop a set of working standards and guidelines

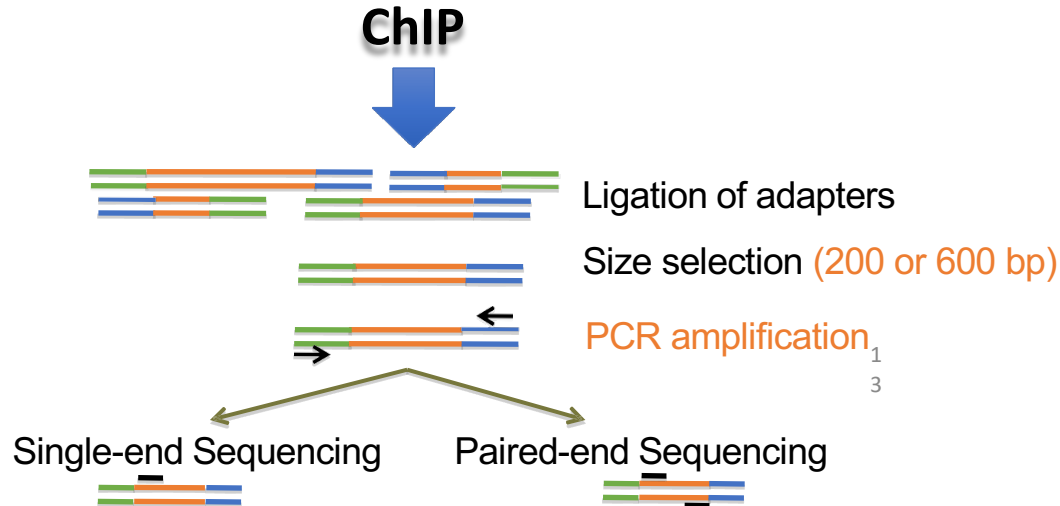


Landt SG, Marinov GK, Kundaje A *et al.* (2012) ChIP-seq guidelines and practices of the ENCODE and modENCODE consortia. *Genome research* **22**, 1813–1831.

See: <https://www.encodeproject.org/about/experiment-guidelines/>

Library prep

- Starting material: ChIP sample (1-10ng of sheared DNA)

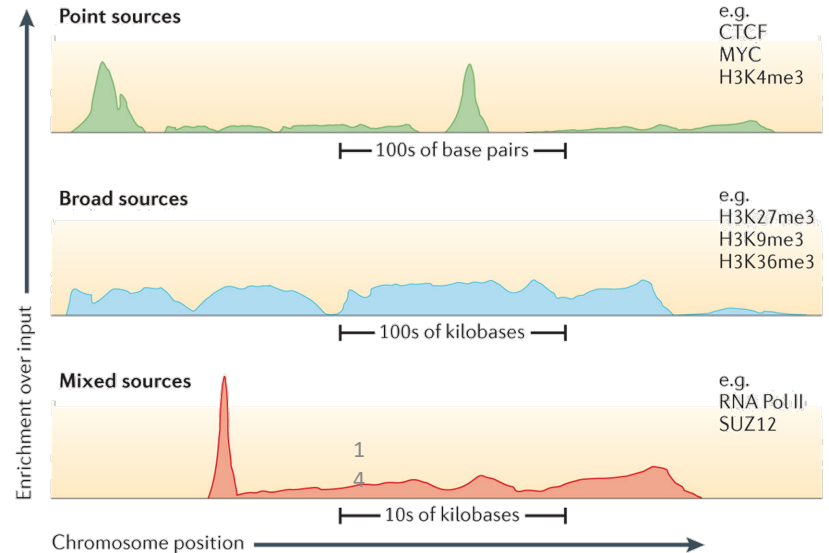


Sequencing

- Paired-end (*single-end*)
- Short reads (50nt)
- Number of reads needed depends on:
 - chipped protein,
 - type of expected profile,
 - number of expected binding sites,
 - size of the genome of interest.

Ex:

- For human genomes, 20 million uniquely mapped read sequences are suggested for point-source peaks, or 40 million for broad-source peaks.
- For fly genome: 8 million reads
- For worm genome: 10 million reads



Nature Reviews | Genetics

Park, 2009

Controls

- Used mostly to filter out false positives (high level of noise)
 - Idea: potential false positive will be enriched in both treatment and control.
- A control will fail to filter out false positives if its enrichment profile is very different from the enrichment profile of false positive regions in the treatment sample
- Most commonly used control: Input DNA (a portion of DNA sample removed prior to IP)
- Choice of control is extremely important
- It is recommended to cover the control in a higher extent than the IPs

Why an Input is required ?

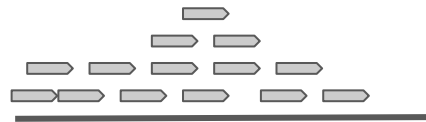
- The input is used to model local noise level
 - Accessible regions are expected to produce more reads



Closed Open Closed

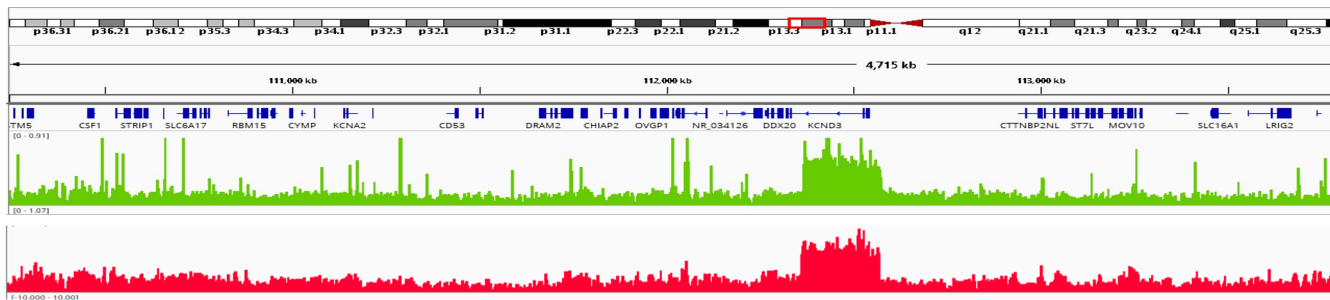


Closed Open Closed



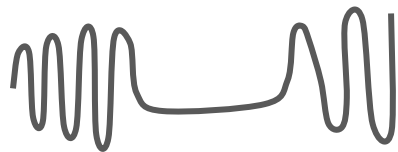
Closed Open Closed

- Amplified regions (CNV) are expected to produce more reads



Why an Input is required ?

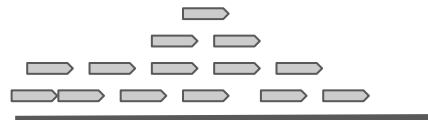
- The input is used to model **local noise level**
 - Accessible regions are expected to produce more reads



Closed Open Closed



Closed Open Closed



Closed Open Closed

- Amplified regions (CNV) are expected to produce more reads
- Moreover, most peak callers are configured with an input as control

Other controls

- IgG (mock IP): controls for non-specific IP enrichment
 - Problem : low-complexity library (few reads)
- Histone H3 (for H3 variants)
- Uninduced condition (for inducible TFs)
 - Example : Glucocorticoid Récepteur
 - Induced by Dexamethasone (Dex)
 - Control vehicule = Ethanol (EthOH)
- KO of your protein of interest
- Non flagged cell lines
- ...



Replicates

- A minimum of **two** replicates should be carried out per experiment.
- Each replicate should be a **biological** rather than a technical replicate; that is, it results from an independent cell culture, embryo pool or tissue sample.

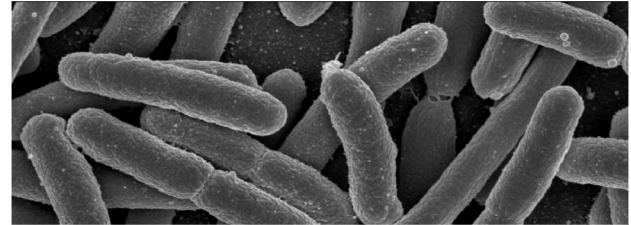


Data analysed in this course

Dataset used

Genome-scale Analysis of *Escherichia coli* FNR Reveals Complex Features of Transcription Factor Binding

Kevin S. Myers^{1,2}, Huihuang Yan^{3aa}, Irene M. Ong³, Dongjun Chung^{4ab}, Kun Liang^{4,5ac}, Frances Tran^{6ad}, Sündüz Keleş^{4,5}, Robert Landick^{3,6,7*}, Patricia J. Kiley^{2,3*}



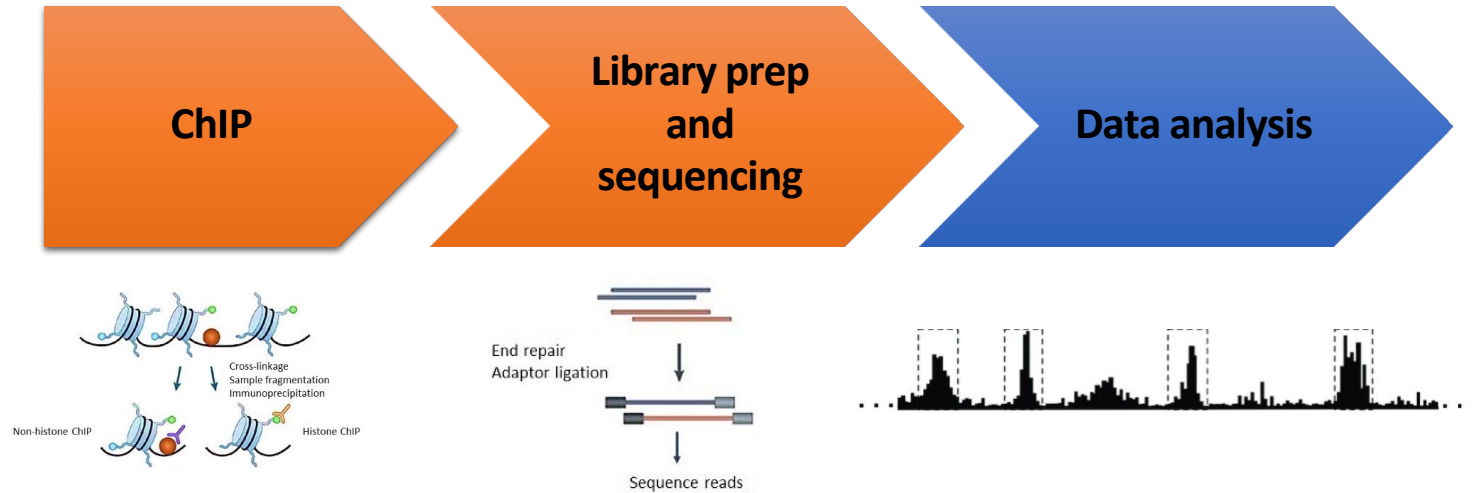
- All experiments (GEO): GSE41187
- Experiment: FNR IP ChIP-seq Anaerobic A (SRX189773 - SRR576933)
- Control: anaerobic INPUT DNA (SRX189778 - SRR576938)



TIPS

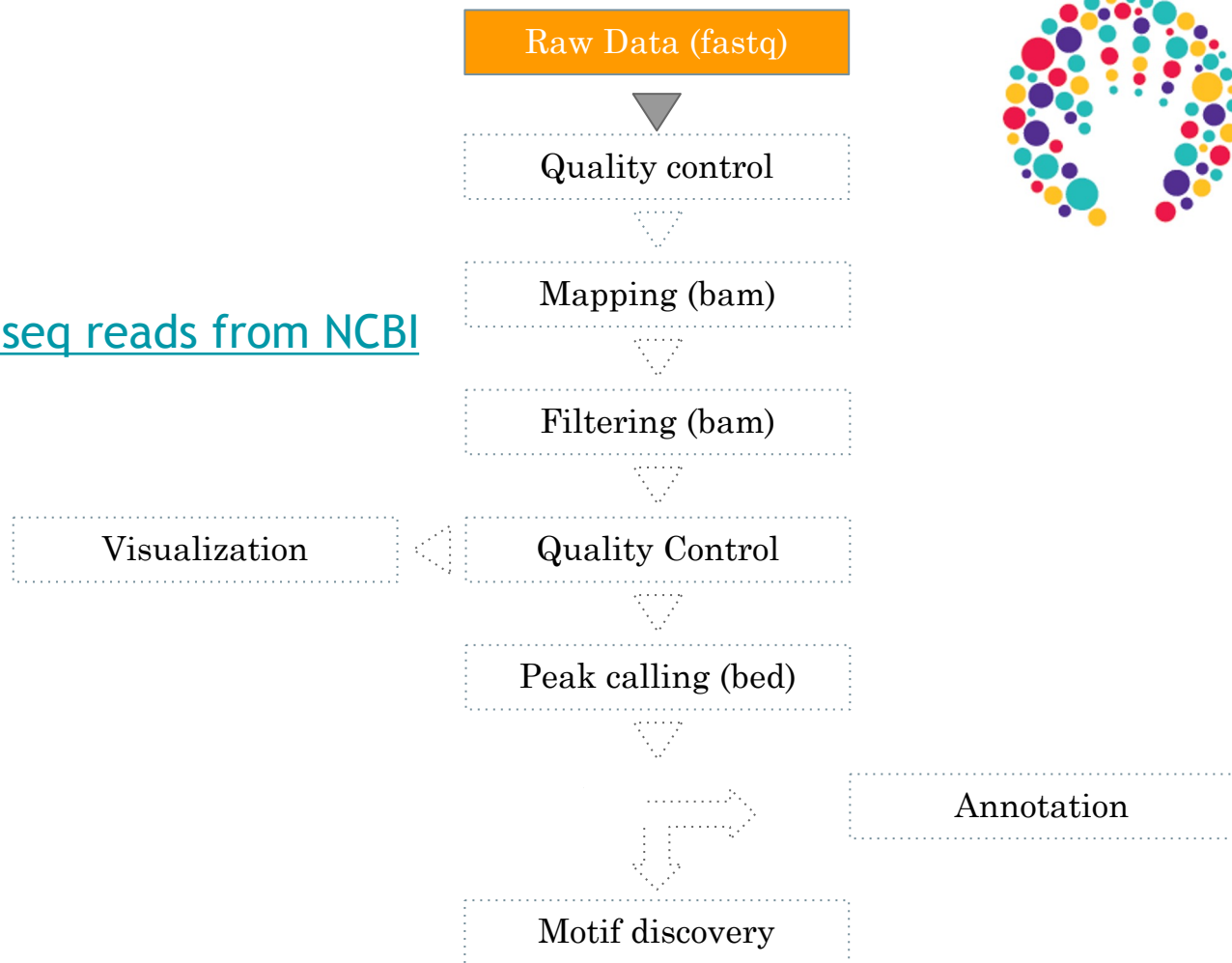
- **Keep track** of all command lines you run. You can for example, create a text file in which you write every commands you run.
- Give **content-explicit names** to the files you're generating.
- Give to files the **right extension**.
- **Create directories with explicit names!!**
- **Compress big files** (with gzip for instance).

Analysis of ChIP-seq data



Protocol

- [Downloading ChIP-seq reads from NCBI](#)



Protocol



- [Connect to the server and set up your environment](#)



Quality control of the reads

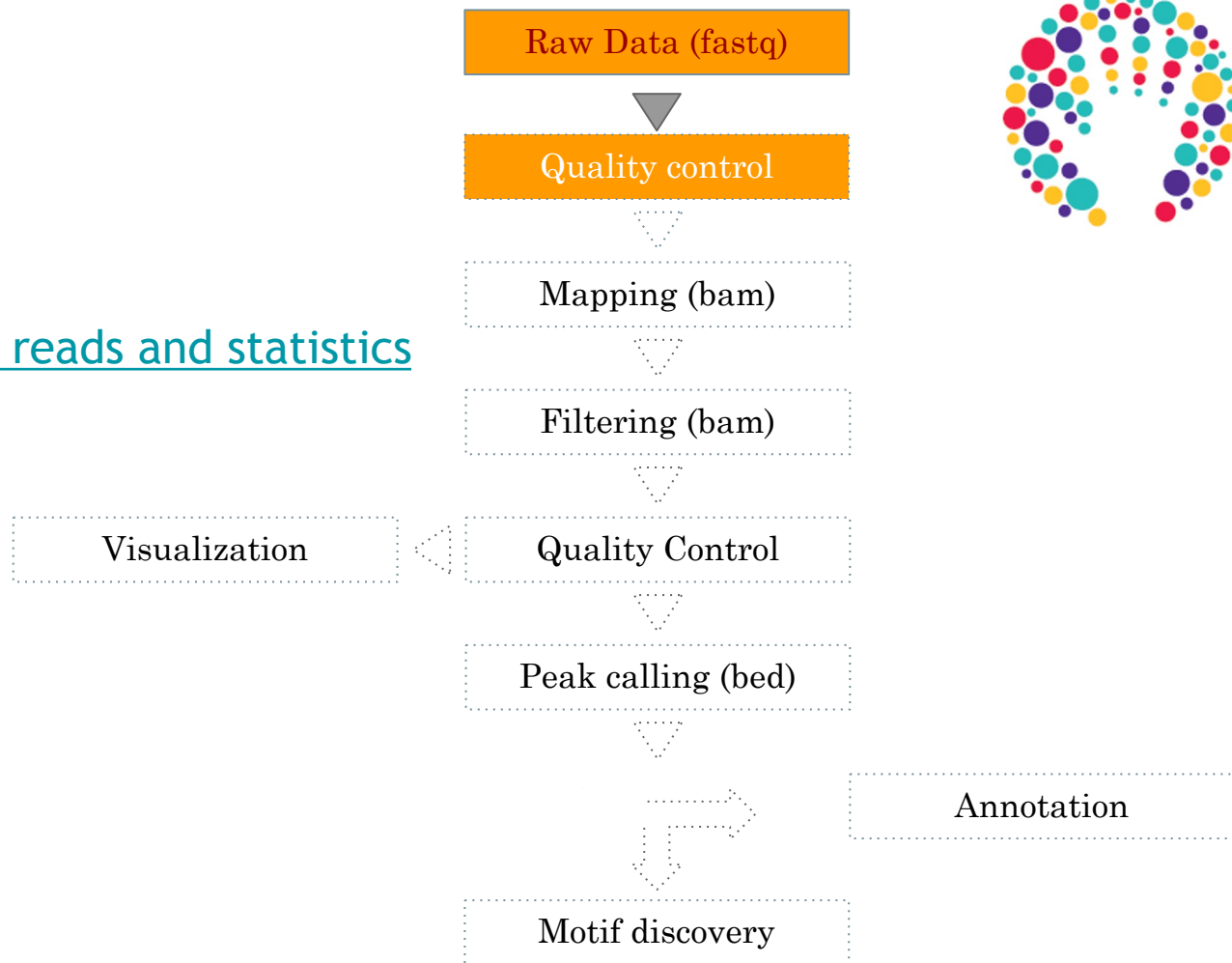


Quality control of the reads

- As for any NGS datasets
- FastQC program

Protocol

- Quality control of the reads and statistics

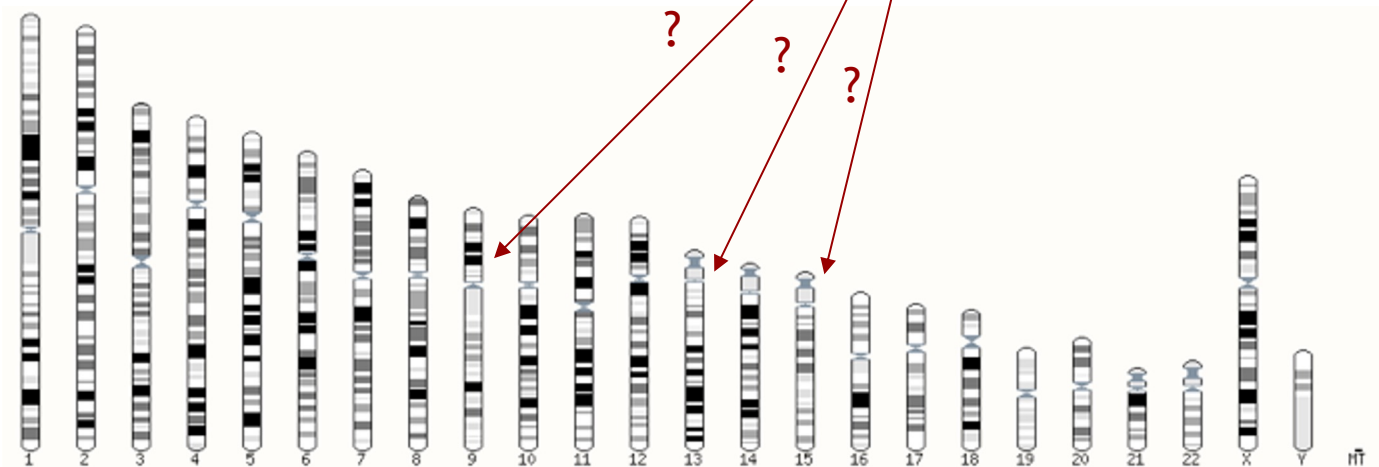




Mapping

Mapping example

ATGCGATTA



Human chromosomes

Genomic coordinates

ATGCGATTA

alignment

1

125,326,318

start

125,326,358

end

Genomic coordinate of the mapped read :

chr5 125326318 125326358 +

181,538,259

5

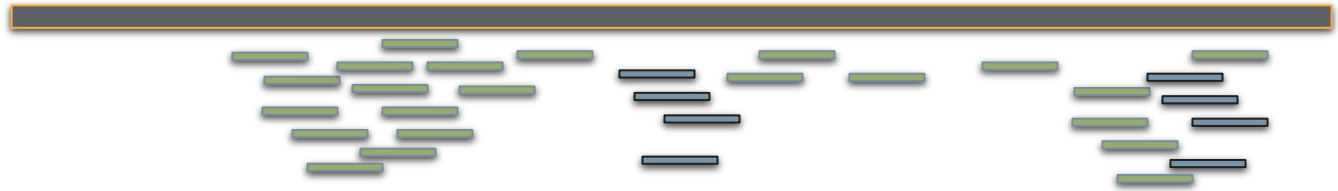
chromosome 5



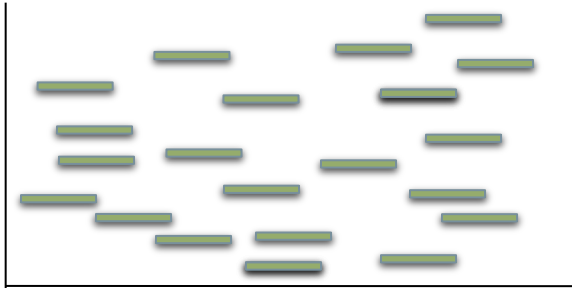
Mapping

- Find out the position of the reads within the reference genome

Ref. Genome



Reads



- One position in the genome
- Many possible positions (Repeat regions, duplicate regions, pseudogenes...)

Mapping tool used: Bowtie

- Designed to align reads if:
 - many of the reads have at least one good, valid alignment,
 - many of the reads are relatively **high-quality**
 - the number of alignments reported per read is small (close to 1)
- Langmead B. et al, Genome Biology 2009
- Langmead B (2010) Aligning short sequencing reads with Bowtie. Curr Protoc Bioinformatics Chapter 11: Unit 11 17

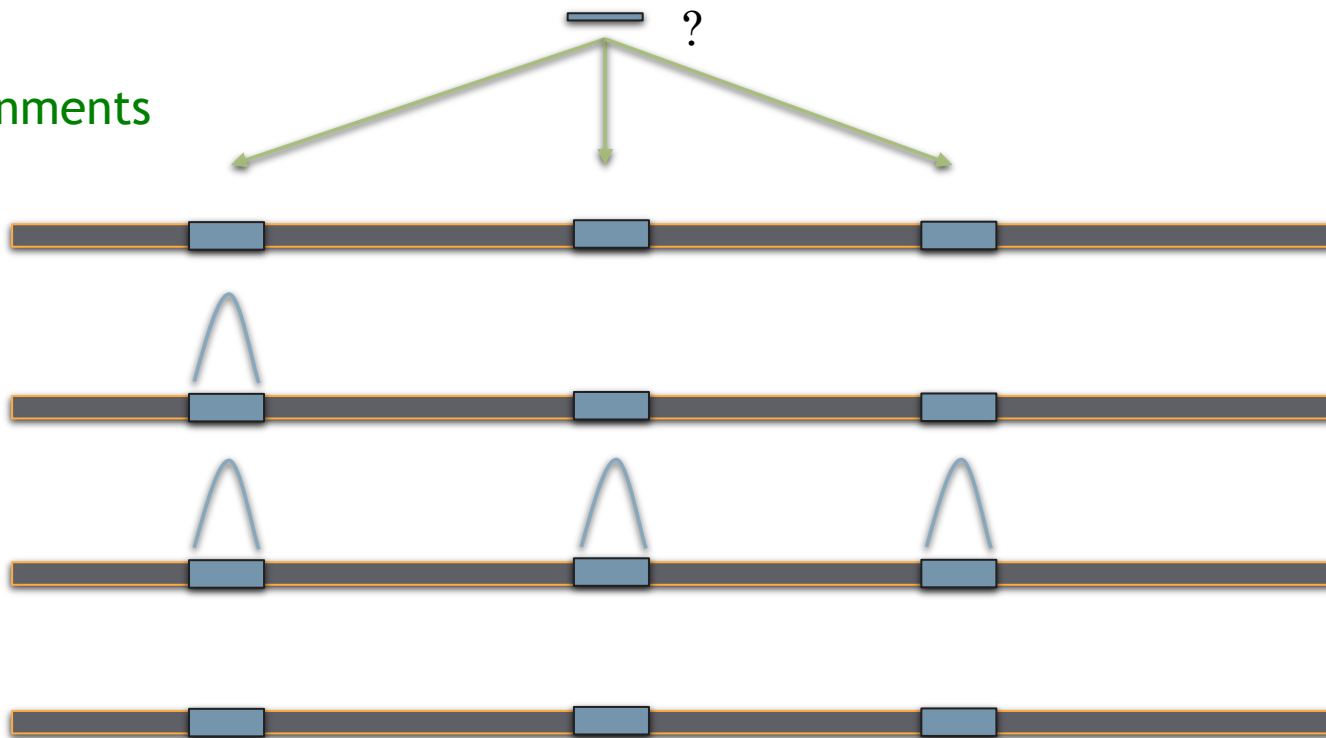
Duplicated genomic regions

3 possible alignments

Keep 1 position
randomly

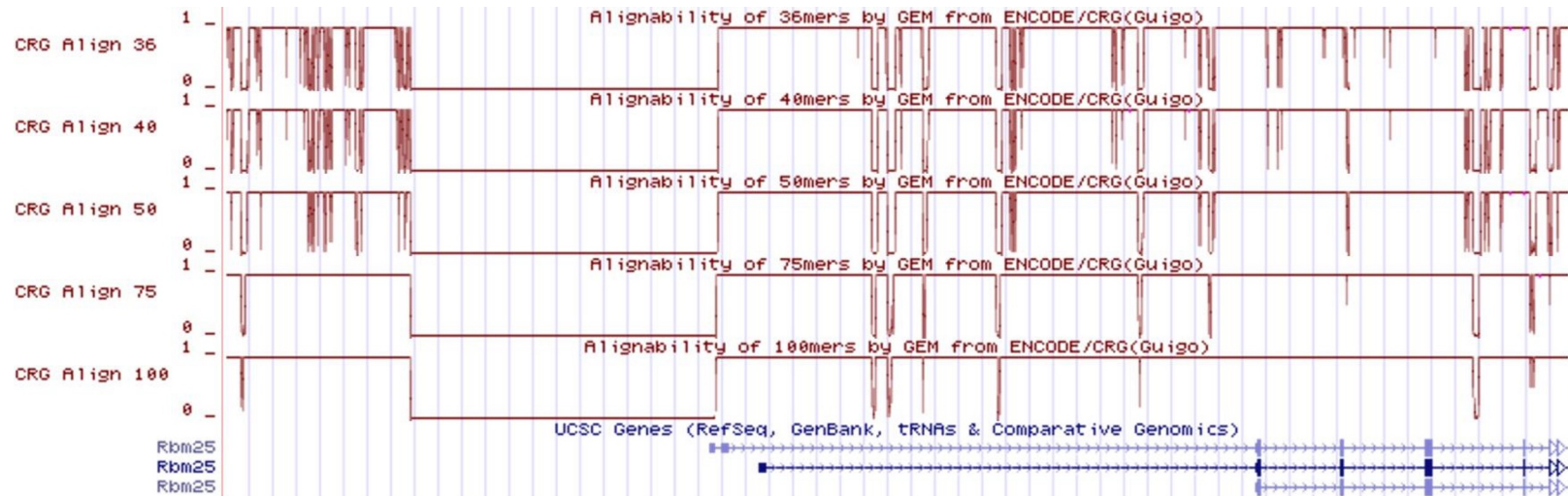
Keep all
possible position

Keep none



Mappability

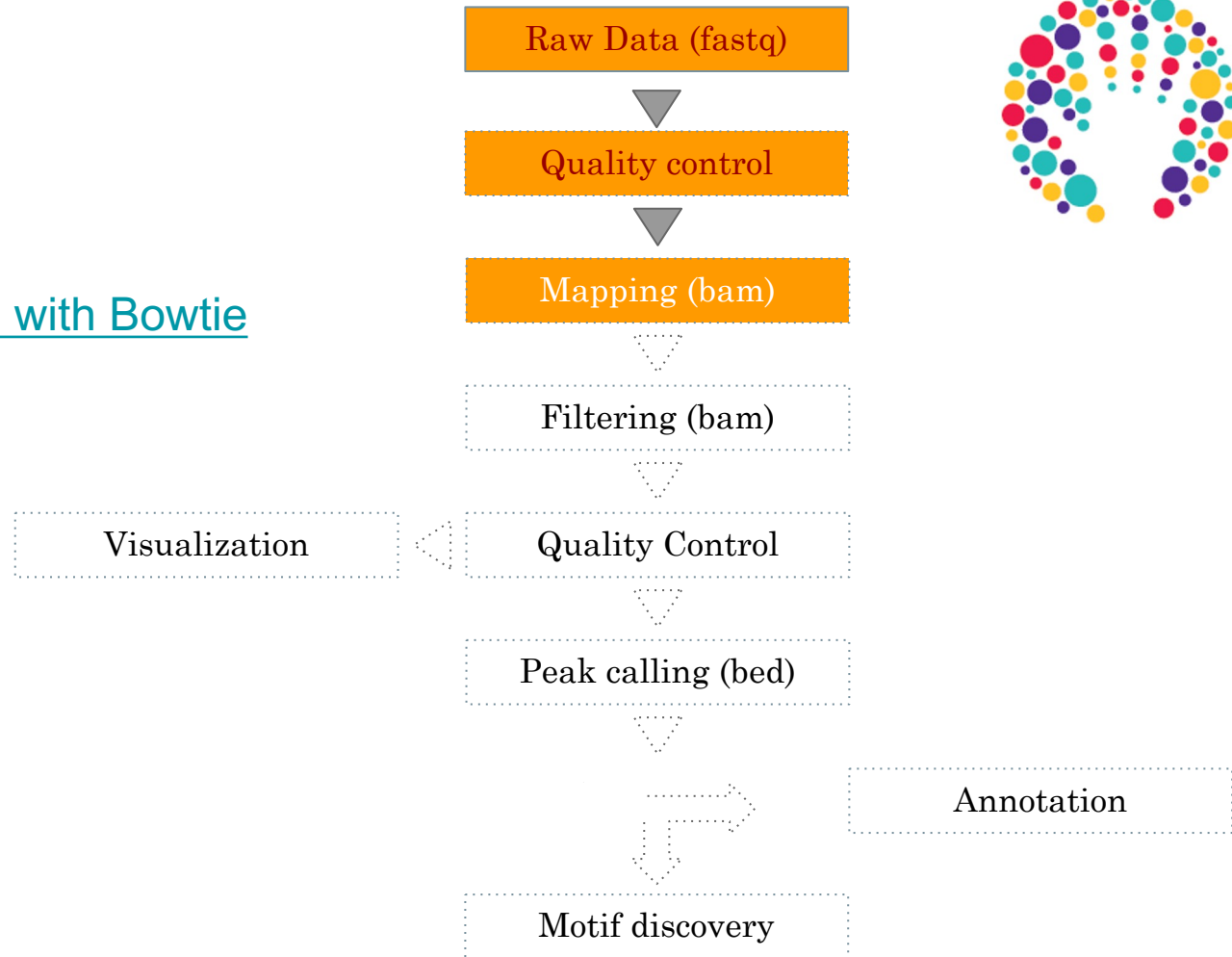
- Mappability (a): how many times a read of a given length can align at a given position in the genome
 - $a=1$ (read align once)
 - $a=1/n$ (read align n times)



Protocol

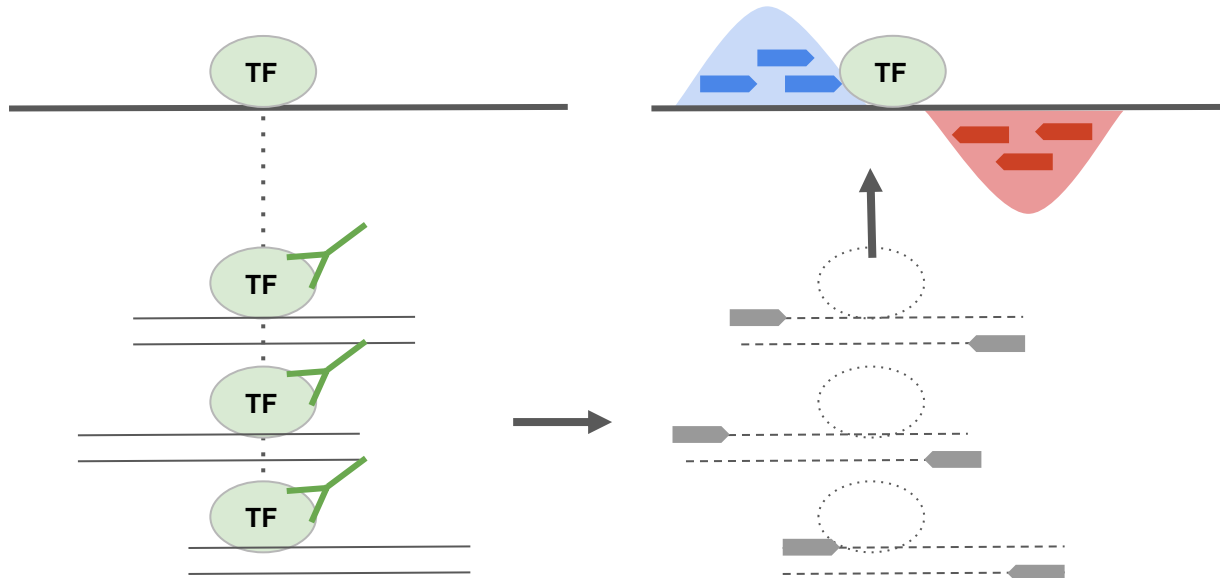


- Mapping the reads with Bowtie



Mapping: expected signal

- For a transcription factor signal is expected to be sharp

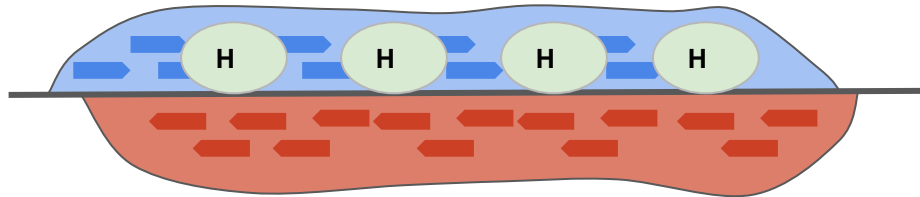


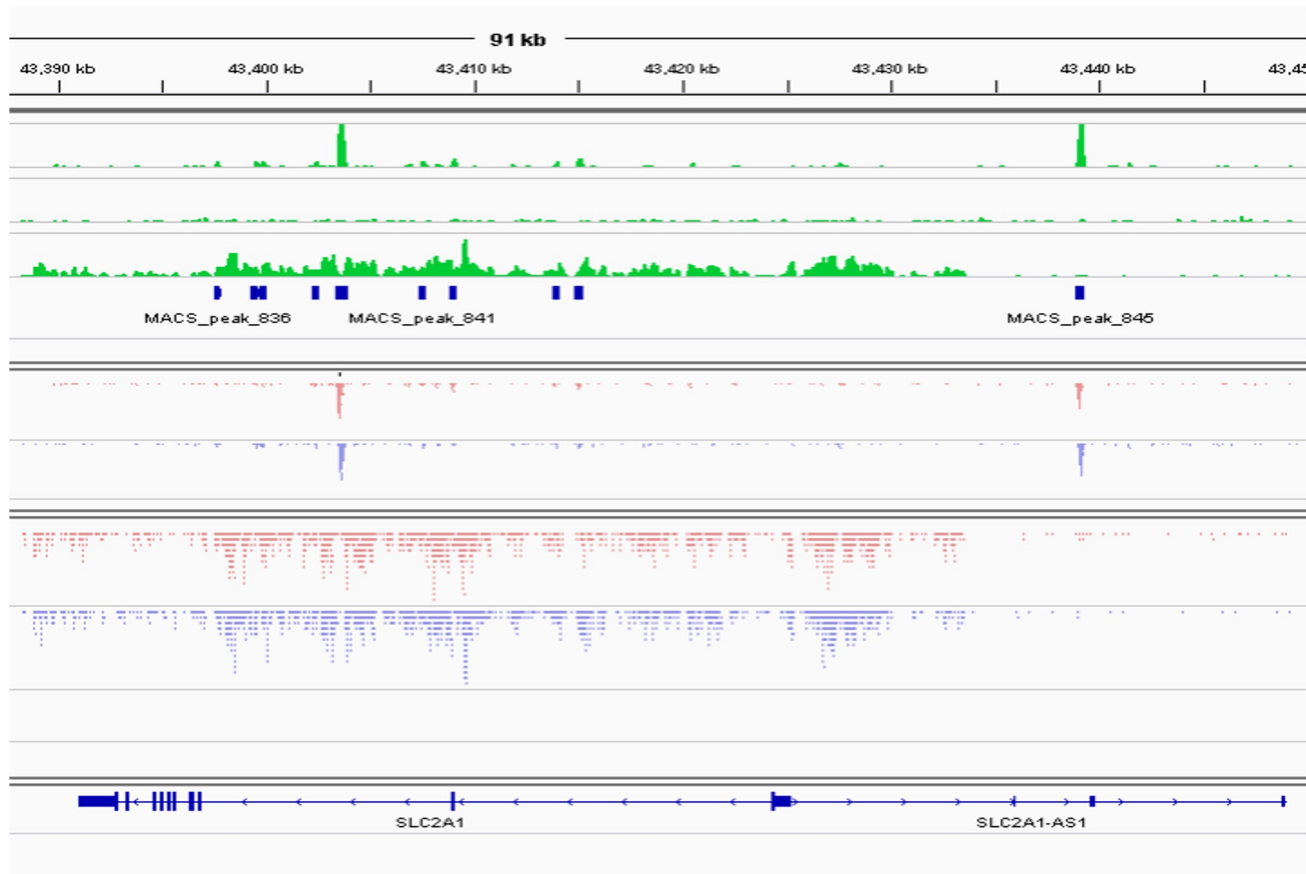
The binding site itself is generally not sequenced !

■ Sens alignments
■ Rev/comp alignments

Mapping: the expected signal

- For most **histone marks** the signal is expected to be **broad**
- Asymmetry is less/not pronounced
- Peak calling algorithms need to adapt to these various signals





Trans. Factor
(ESR1)

Histone mark
(H3K4me1)



Filtering mapped reads

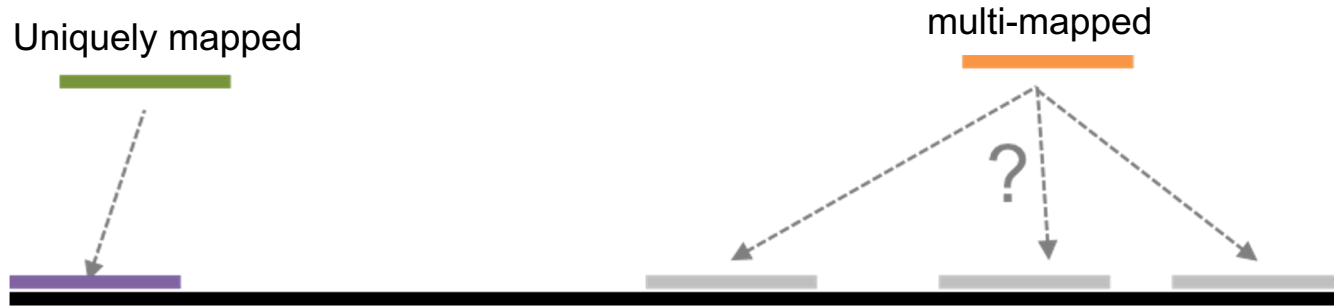


Which reads to filter ?

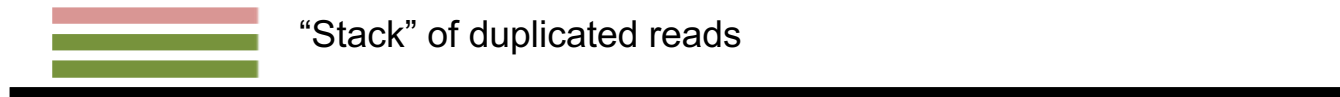
- Low-quality read alignments
 - Tool : samtools
- Multi-mapped reads (unless removed during the mapping step)
 - Tool : samtools
- Duplicated reads (PCR duplicates)
 - Tool : Picard MarkDuplicates

Source of confusion

uniquely mapped reads = reads that “matches” only 1 region in the genome



duplicate reads = reads that “match” at the SAME location (same start, strand)

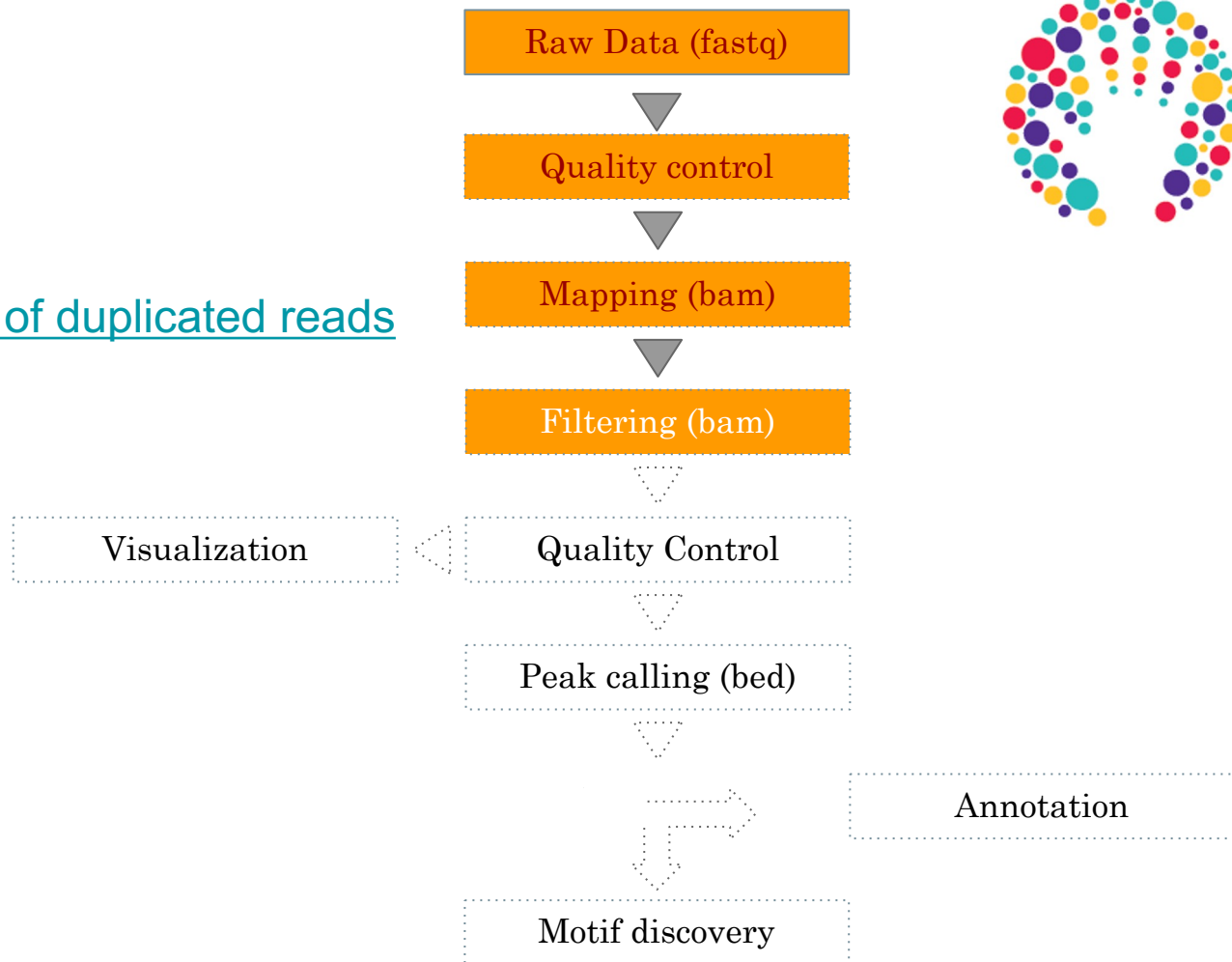


PCR duplicates

- PCR duplicates
 - Related to poor library complexity
 - The same set of fragments are amplified, may indicate that immuno-precipitation failed
 - Tools to check for
 - FastQC report (duplicate diagram)
 - PCR bottleneck metric (ENCODE)

Protocol

- Estimating the number of duplicated reads





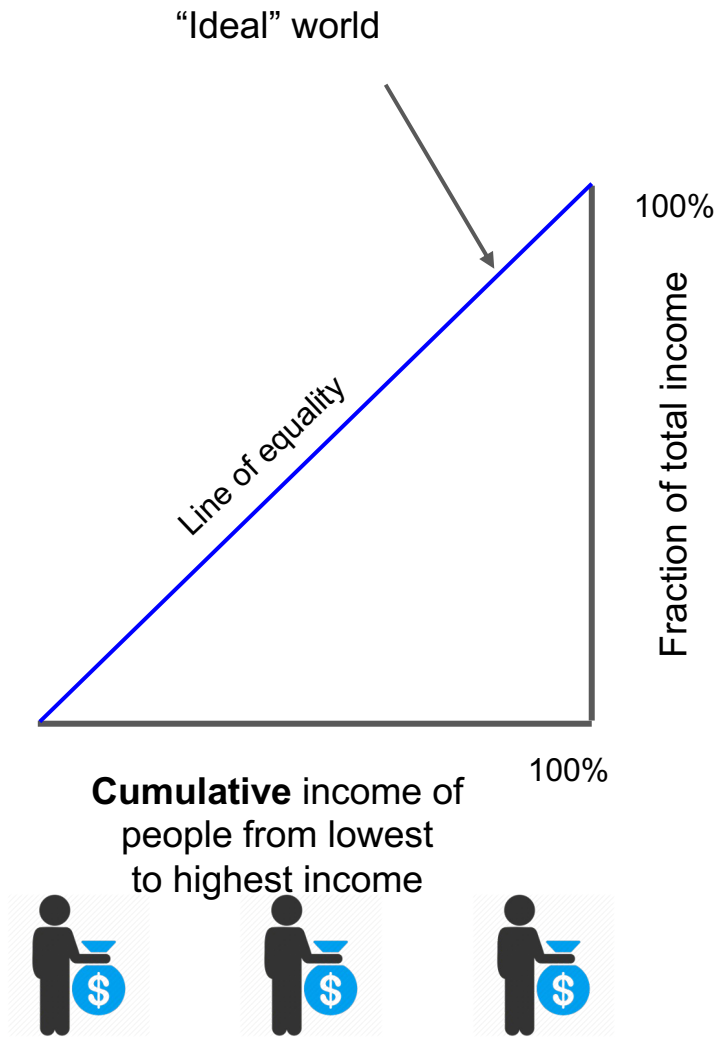
Quality Control on mapped reads

Assessing ChIP quality

- Guidelines from ENCODE
- Various metrics
 - Check **duplicate** rate (see previous Filtering section)
 - Use a **Lorenz Curve** (implemented in **Deeptools fingerprint**)
 - Look at **strand cross-correlation** (implemented in **SPP BioC package** and **phantompeakqualtools**)
 - Fraction of reads in peaks (**FRiP**, as proposed by ENCODE), but requires to find peaks.

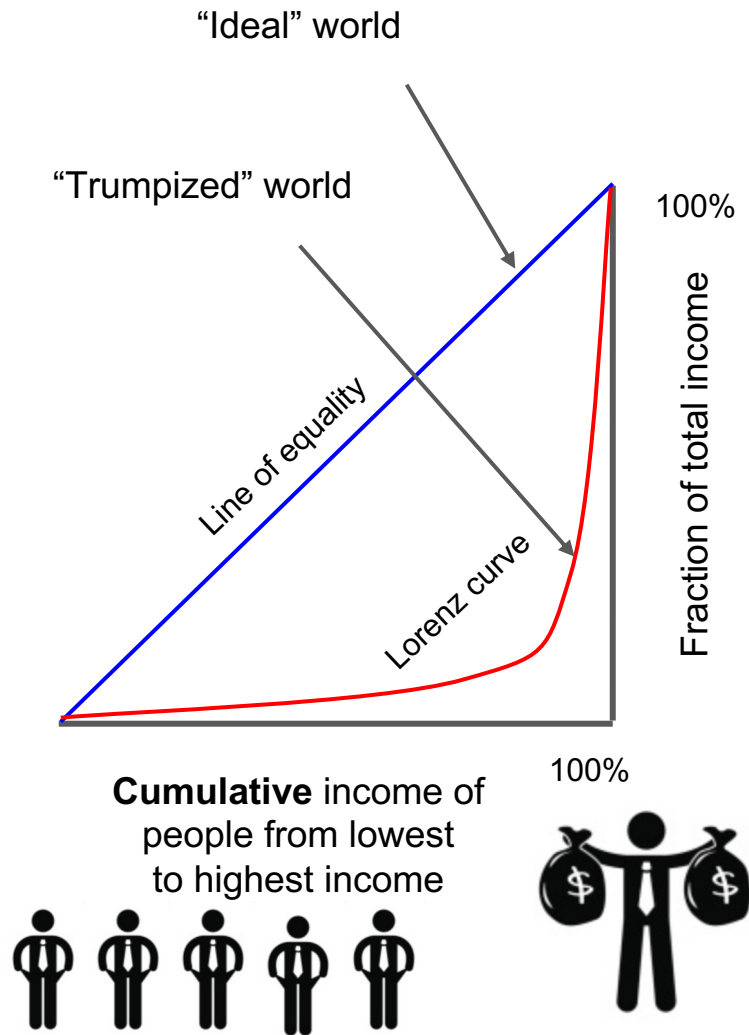
Lorenz curve

- Analyze income among workers by computing cumulative sum.
 - If uniform income distribution :
 - Straight line



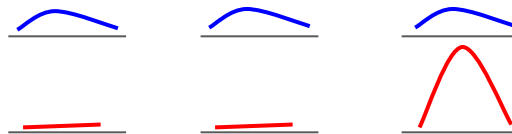
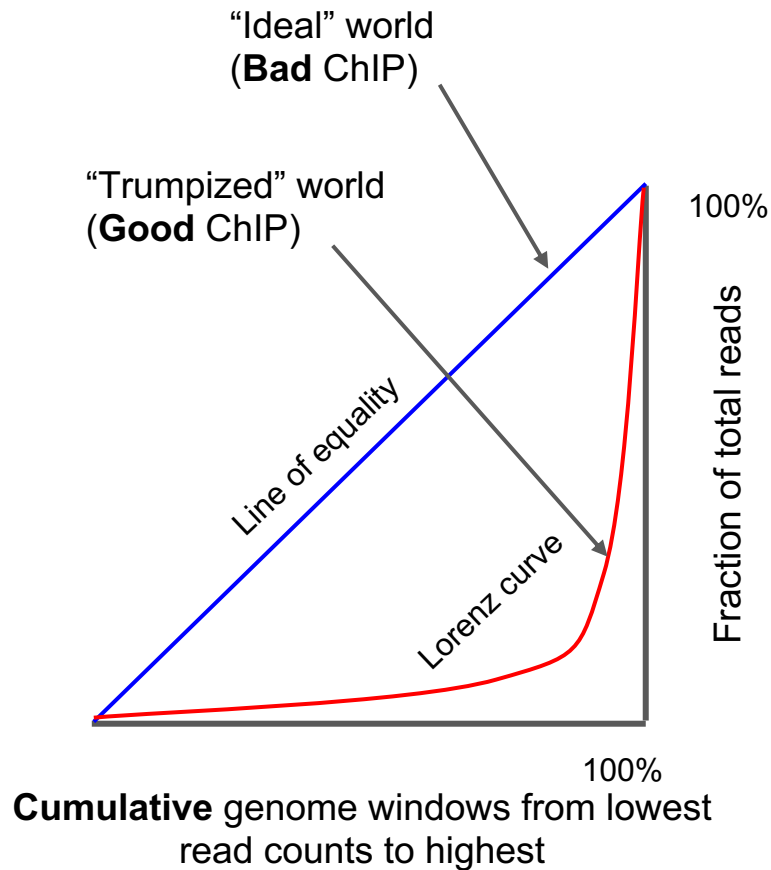
Lorenz curve

- Analyze income among workers by computing cumulative sum.
 - If uniform income distribution :
 - **Straight line**
 - If they were trumpized
 - **Lorenz curve**



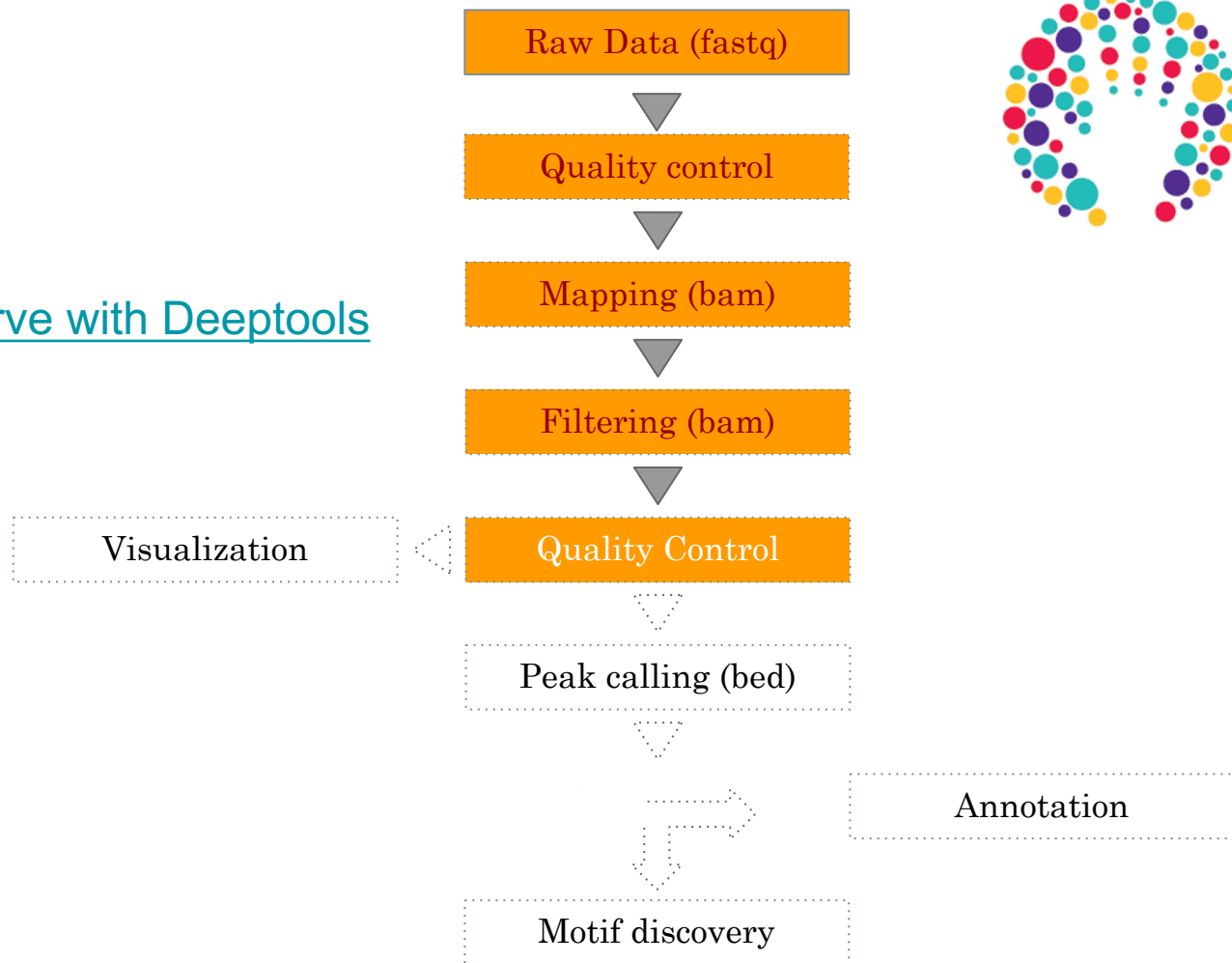
Lorenz curve

- Analyze income among workers by computing cumulative sum.
 - If uniform income distribution :
 - **Straight line**
 - If they were trumpized
 - **Lorenz curve**
- Here the workers are the genome windows and incomes are reads



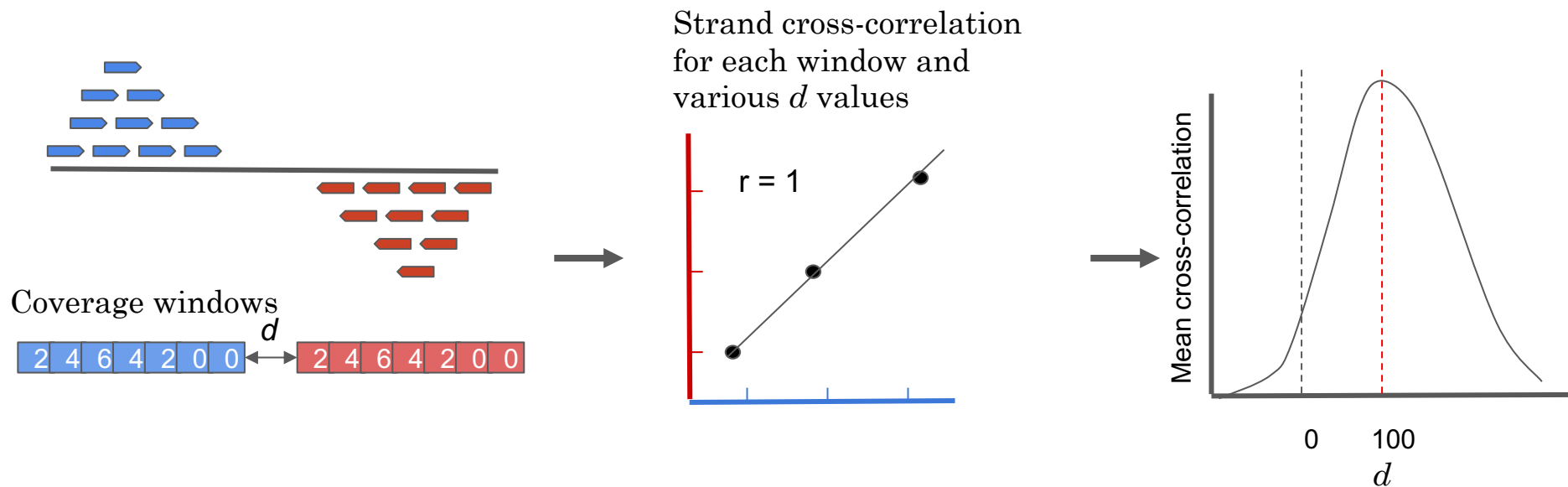
Protocol

- Plot the Lorenz curve with Deeptools

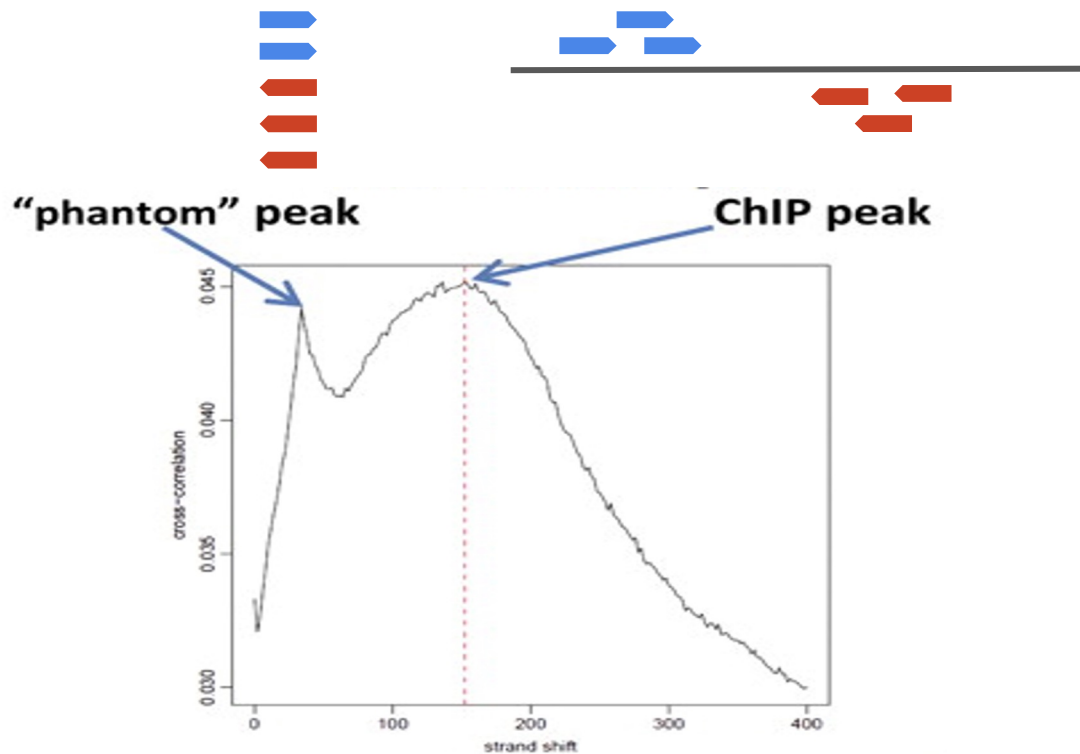


Strand cross-correlation

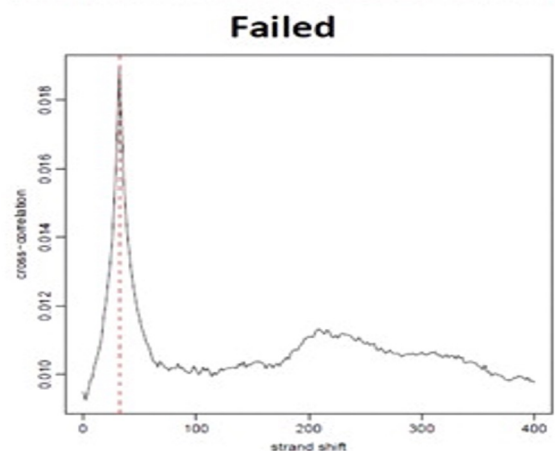
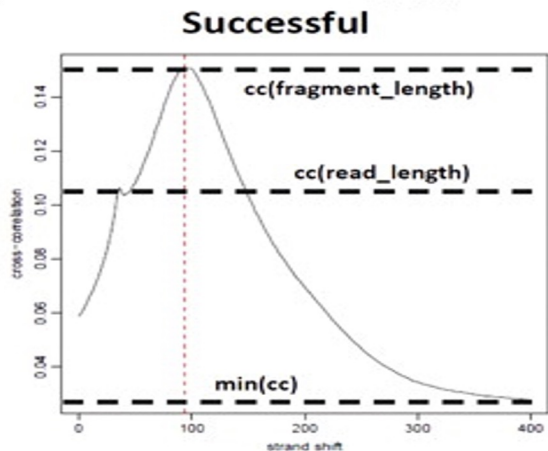
- Compute strand cross correlation for each window w across the genome.
- Use various distance d and compute the mean cross-correlation observed



Strand cross-correlation



Strand cross-correlation



NSC: normalized strand coefficient

$$NSC = \frac{cc(fragment\ length)}{min(cc)}$$

NSC $\geq$ 1.05 is recommended

Relative strand correlation (RSC)

$$RSC = \frac{cc(fragment\ length) - min(cc)}{cc(read\ length) - min(cc)}$$

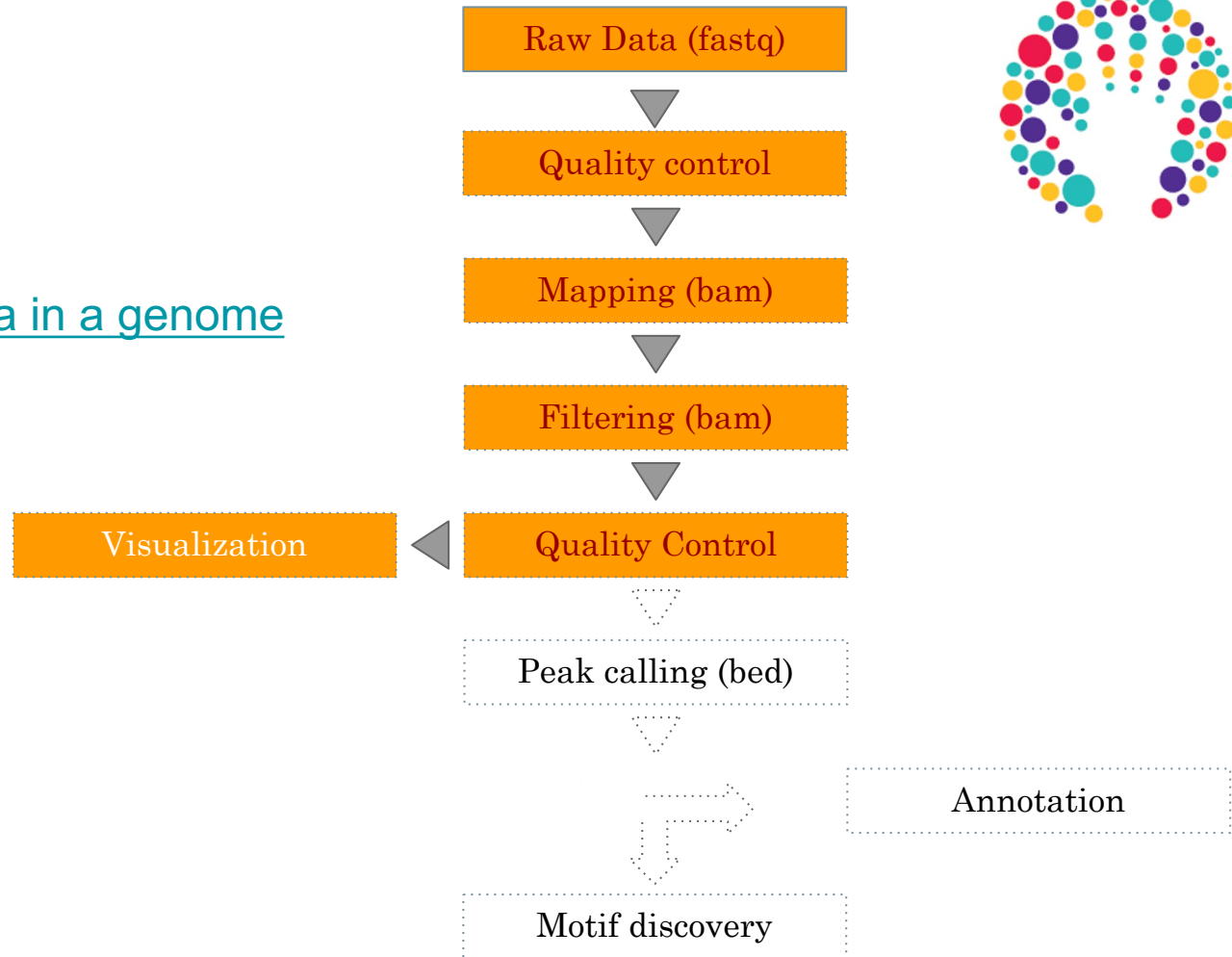
RSC $\geq$ 0.8 is recommended



Visualization: computing a genomic
coverage file

Protocol

- [Visualizing the data in a genome browser](#)

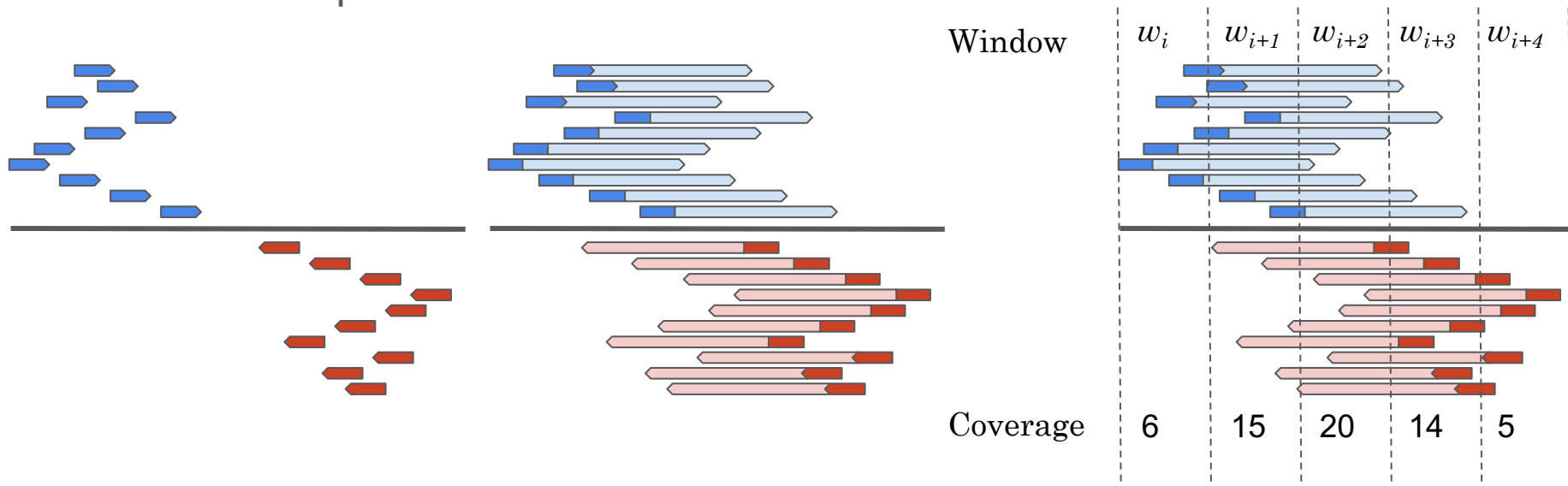


Bam files are fat

- **BAM files are fat** as they do contain exhaustive information about read alignments
 - Memory issues (can only visualize fraction of the BAM)
- Need a more **lightweight** file format containing **only genomic coverage** information:
 - ❌ Wig (not compressed, not indexed)
 - ✅ TDF (compressed, indexed)
 - ✅ BigWig (compressed, indexed)

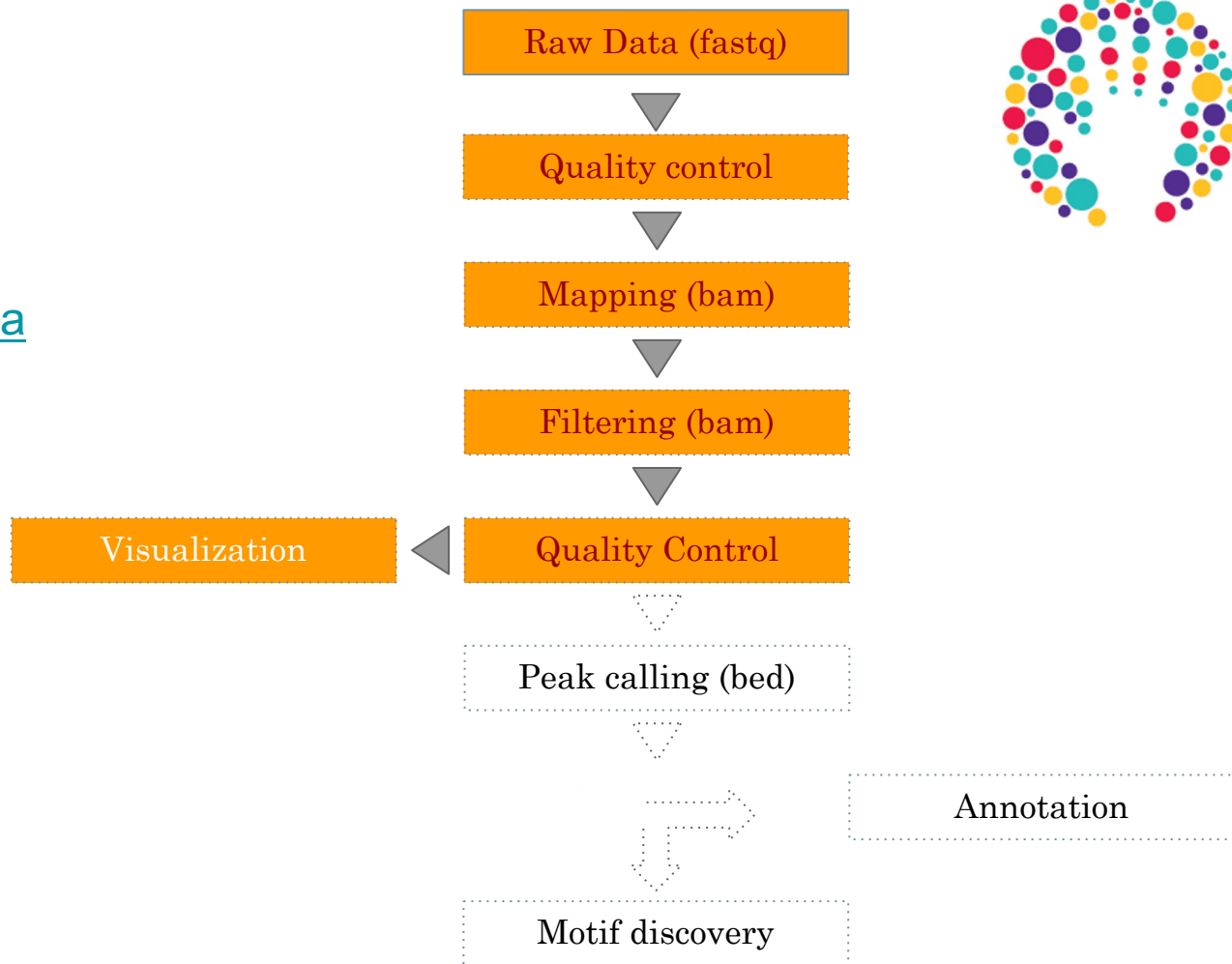
Coverage file and read extension

- BAM files do not contain fragment location but read location
- We need to extend reads to compute fragments coordinates before coverage analysis
- Not required for PE



Protocol

- [Viewing scaled data](#)

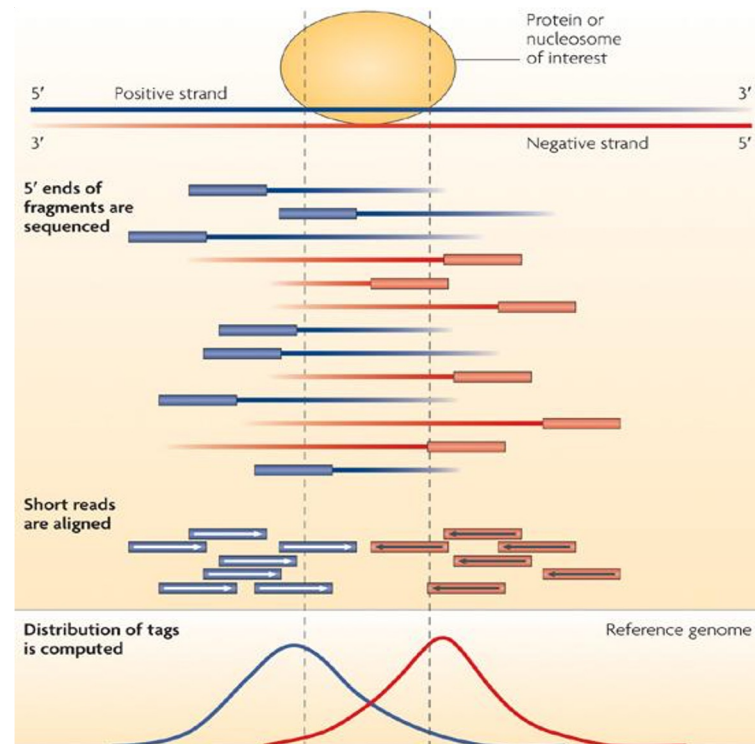




Peak Calling

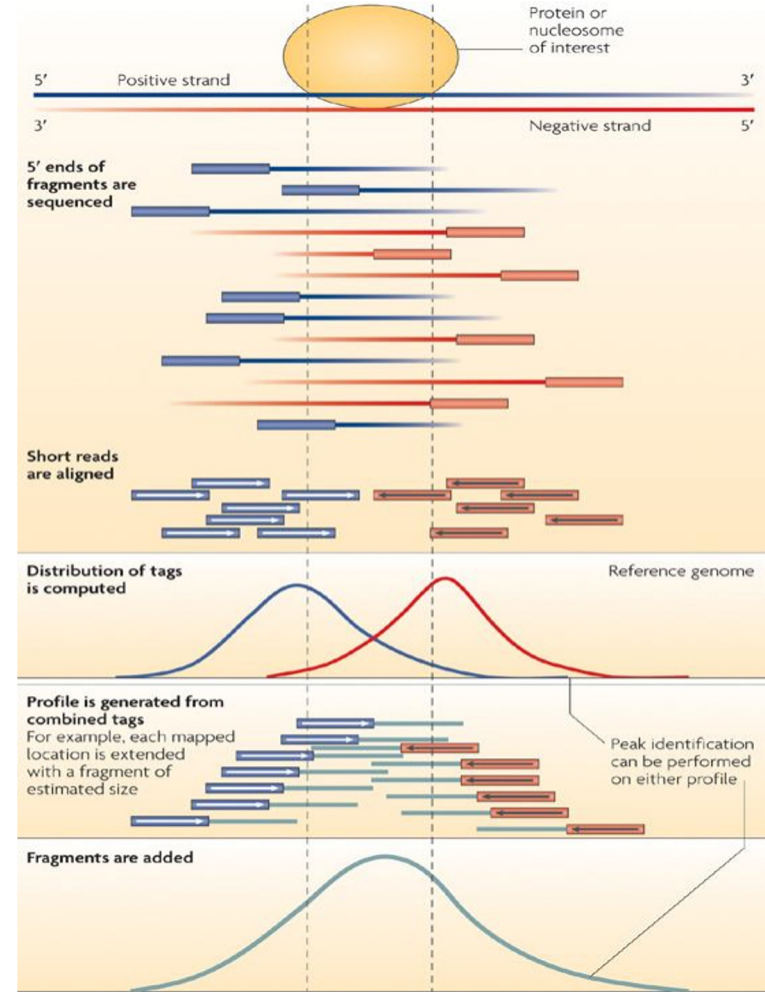
From reads to peaks

- Chip-seq peaks are a mixture of two signals:
 - + strand reads (Watson)
 - - strand reads (Crick)
- The sequence read density accumulates on forward and reverse strands centered around the binding site



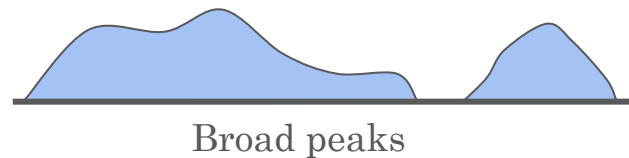
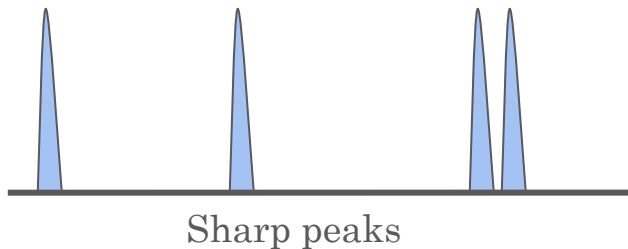
From reads to peaks

- Get the signal at the right position
 - Read shift
 - Extension
- Estimate the fragment size
- Do paired-end



Peak callers

- The peak caller should be chosen based on
 - Experimental design
 - SE or PE (E.g MACS1.4 vs MACS2)
 - Expected signal
 - Sharp peaks (e.g. Transcription Factors).
 - E.g. MACS
 - Broad peaks (e.g. epigenetic marks).
 - E.g. MACS, SICER,...



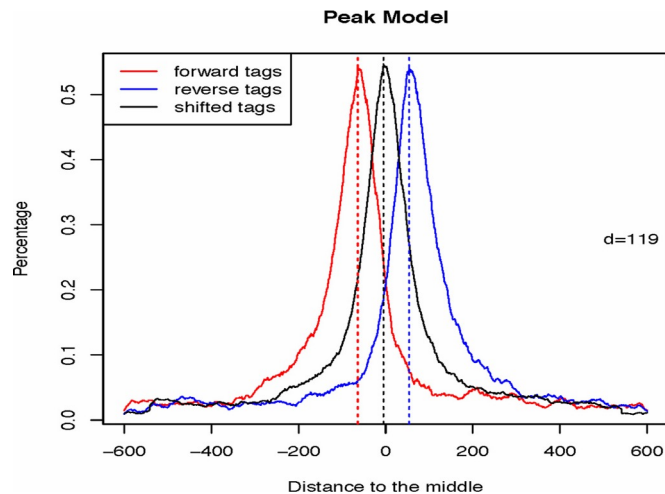
Peak callers

	Profile	Peak criteria ^a	Tag shift	Control data ^b	Rank by	FDR ^c	User input parameters ^d	Artifact filtering: strand-based/duplicate ^e	Refs.
CisGenome v1.1	Strand-specific window scan	1: Number of reads in window 2: Number of ChIP reads minus control reads in window	Average for highest ranking peak pairs	Conditional binomial used to estimate FDR	Number of reads under peak	1: Negative binomial 2: conditional binomial	Target FDR, optional window width, window interval	Yes / Yes	10
ERANGE v3.1	Tag aggregation	1: Height cutoff High quality peak estimate, per-region estimate, or input	High quality peak estimate, per-region estimate, or input	Used to calculate fold enrichment and optionally <i>P</i> values	<i>P</i> value	1: None 2: # control / # ChIP	Optional peak height, ratio to background	Yes / No	4,18
FindPeaks v3.1.9.2	Aggregation of overlapped tags	Height threshold	Input or estimated	NA	Number of reads under peak	1: Monte Carlo simulation 2: NA	Minimum peak height, subpeak valley depth	Yes / Yes	19
F-Seq v1.82	Kernel density estimation (KDE)	<i>s</i> s.d. above KDE for 1: random background, 2: control	Input or estimated	KDE for local background	Peak height	1: None 2: None	Threshold s.d. value, KDE bandwidth	No / No	14
GLTR	Aggregation of overlapped tags	Classification by height and relative enrichment	User input tag extension	Multiply sampled to estimate background class values	Peak height and fold enrichment	2: # control / # ChIP	Target FDR, number nearest neighbors for clustering	No / No	17
MACS v1.3.5	Tags shifted then window scan	Local region Poisson <i>P</i> value	Estimate from high quality peak pairs	Used for Poisson fit when available	<i>P</i> value	1: None 2: # control / # ChIP	<i>P</i> -value threshold, tag length, mfold for shift estimate	No / Yes	13
PeakSeq	Extended tag aggregation	Local region binomial <i>P</i> value	Input tag extension length	Used for significance of sample enrichment with binomial distribution	<i>q</i> value	1: Poisson background assumption 2: From binomial for sample plus control	Target FDR	No / No	5
QuEST v2.3	Kernel density estimation	2: Height threshold, background ratio	Mode of local shifts that maximize strand cross-correlation	KDE for enrichment and empirical FDR estimation	<i>q</i> value	1: NA 2: # control / # ChIP as a function of profile threshold	KDE bandwidth, peak height, subpeak valley depth, ratio to background	Yes / Yes	9
SICER v1.02	Window scan with gaps allowed	<i>P</i> value from random background model, enrichment relative to control	Input	Linearly rescaled for candidate peak rejection and <i>P</i> values	<i>q</i> value	1: None 2: From Poisson <i>P</i> values	Window length, gap size, FDR (with control) or <i>E</i> -value (no control)	No / Yes	15
SiSSRs v1.4	Window scan	$N_c - N_r$ sign change, $N_c + N_r$ threshold in	Average nearest paired tag distance	Used to compute fold-enrichment distribution	<i>P</i> value	1: Poisson 2: control distribution	1: FDR 1,2: $N_c + N_r$ threshold	Yes / Yes	11

MACS [Zhang et al, 2008]

1. Modeling the shift size of ChIP-Seq tags

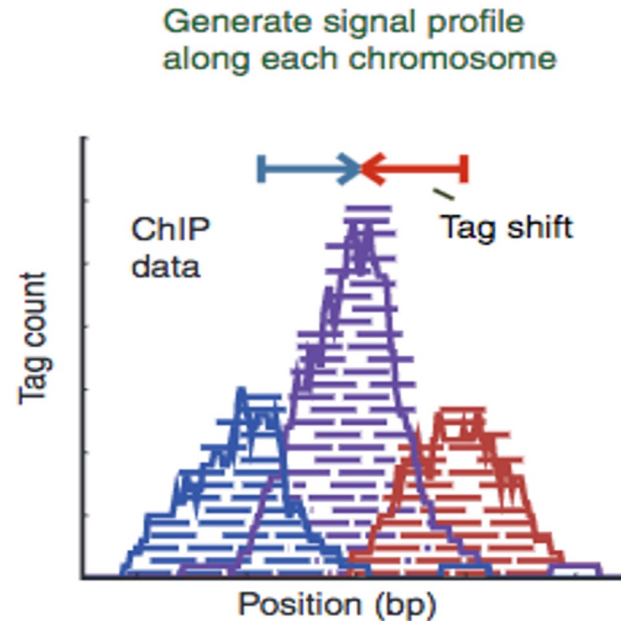
- slides $2 \times \text{bandwidth}$ windows across the genome to find regions with tags more than $m\text{fold}$ enriched relative to a random tag genome distribution
- randomly samples 1,000 of these highly enriched regions
- separates their + and - reads, and aligns them by the midpoint between their + and - read centers
- define d as the distance in bp between the summit of the two distribution



MACS [Zhang et al, 2008]

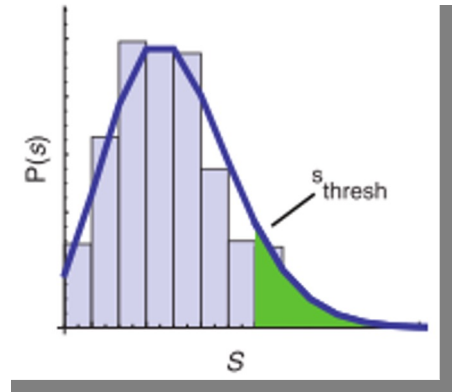
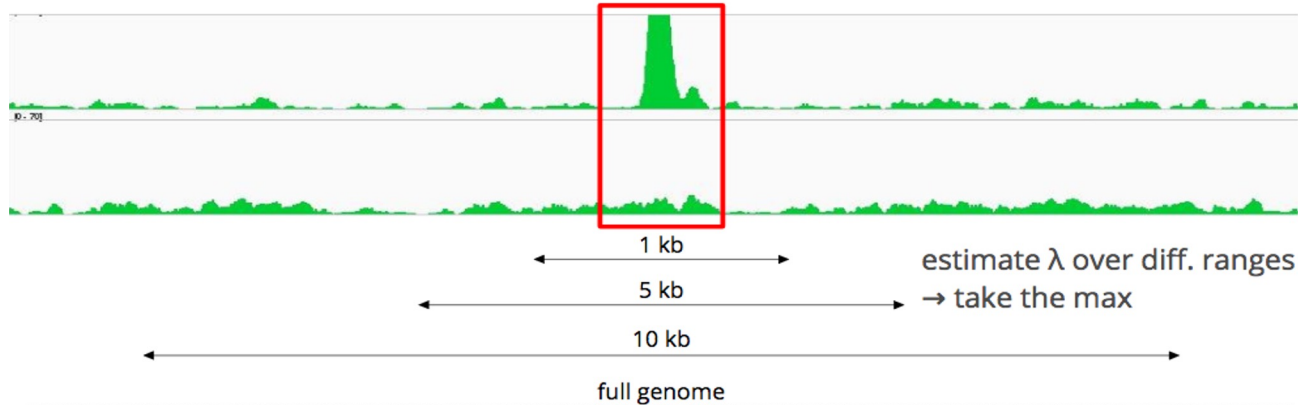
2. Peak detection

- Scales the total Input read count to be the same as the total ChIP read count
- Duplicate read removal
- Reads are shifted by $d/2$
(d value is the model obtained in step 1)



MACS [Zhang et al, 2008]

- Slides 2d windows across the genome to find candidate peaks with a significant read enrichment (Poisson distribution p -value based on λ_{BG} , default 10^{-5})
- Estimate parameter λ_{local} of Poisson distribution
- Keep peaks significant under λ_{BG} and λ_{local} and with p -value < threshold



MACS [Zhang et al, 2008]

3. Multiple testing correction (FDR)

- Swap treatment and input and call negative peaks
- Take all the peaks (neg + pos) and sort them by increasing p-values

$$\text{FDR}(p) = \frac{\text{\# Negative peaks with p-value} < p}{\text{\# Selected peaks}}$$



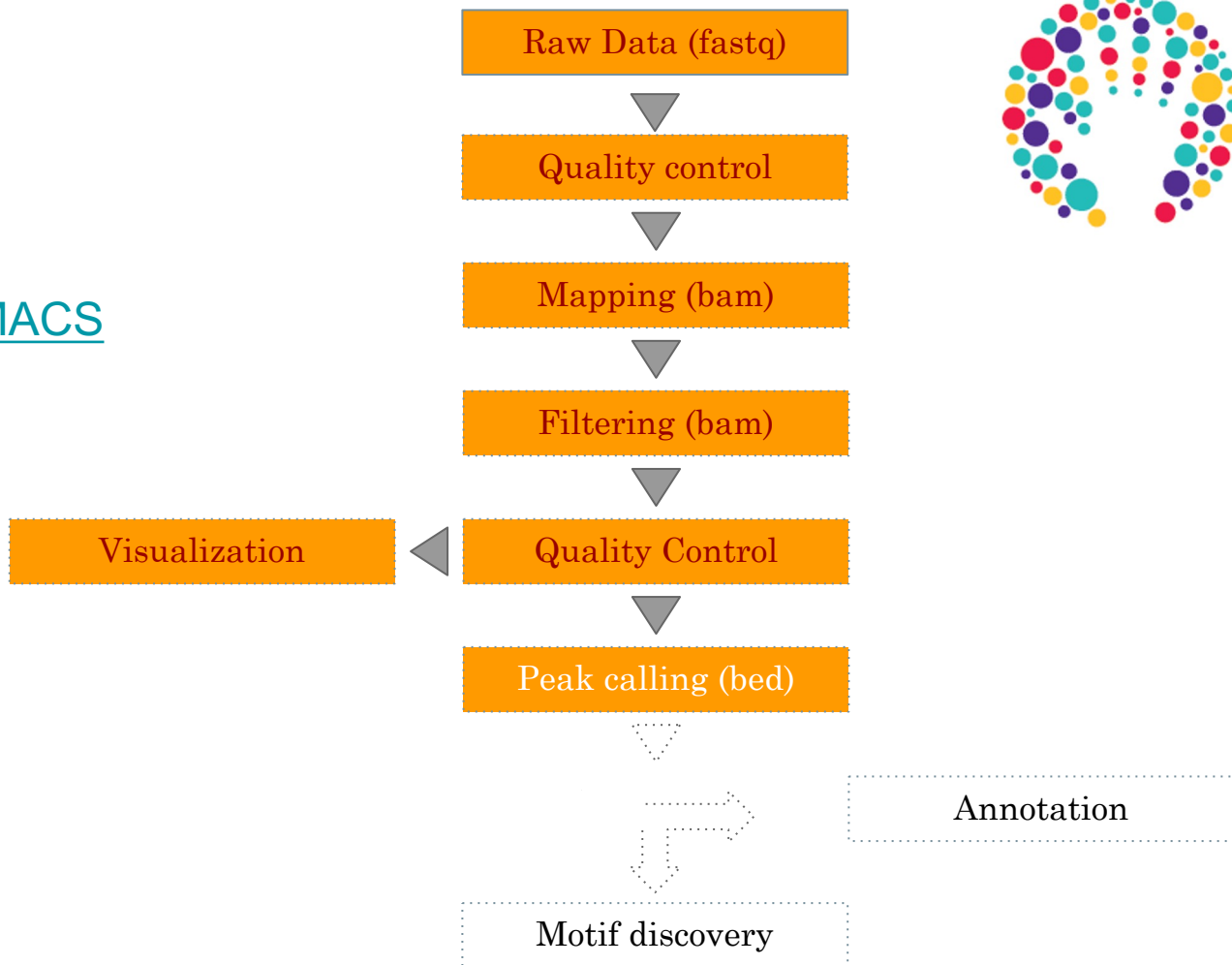
MACS in summary

- Step 1 : search for candidate regions that look like good peaks, to produce a fine-tuned **model** of the peaks (d value) to search in Step 2
- Step 2 : actual peak calling
 - **sliding window** length = $2*d$
 - In each window : test if the region is a peak, by comparing the number of reads in the treatment and the expected number of reads
 - Comparison is based on a **statistical test** with a Poisson distribution, keeping only regions with **p-value < threshold**
- Step 3 : correction for multiple testing (many windows were tested), calculation of **FDR**

Protocol

- Peak calling with MACS

(stop after step 5)

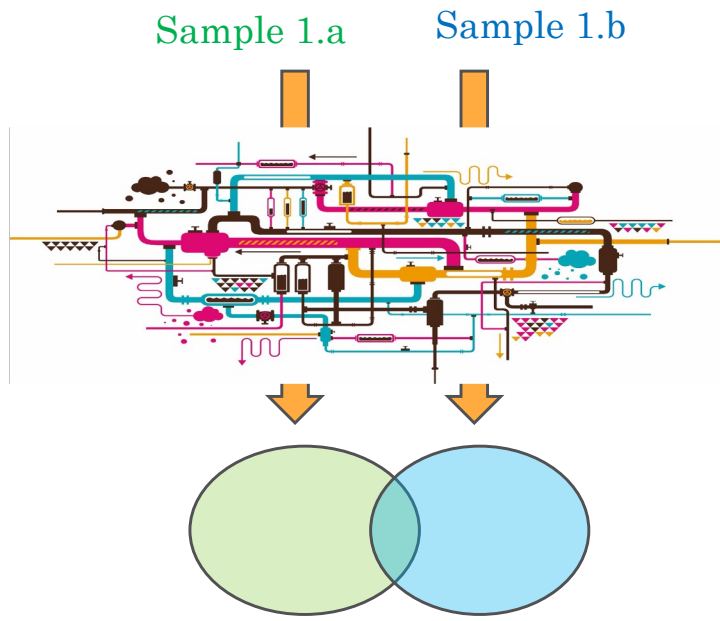




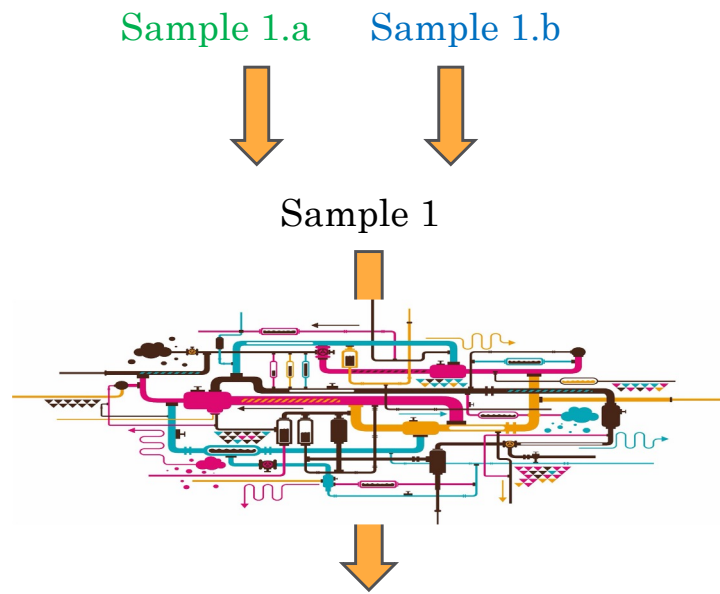
How to deal with replicates

How to deal with replicates

Analyze samples separately and takes union or intersection of resulting peaks



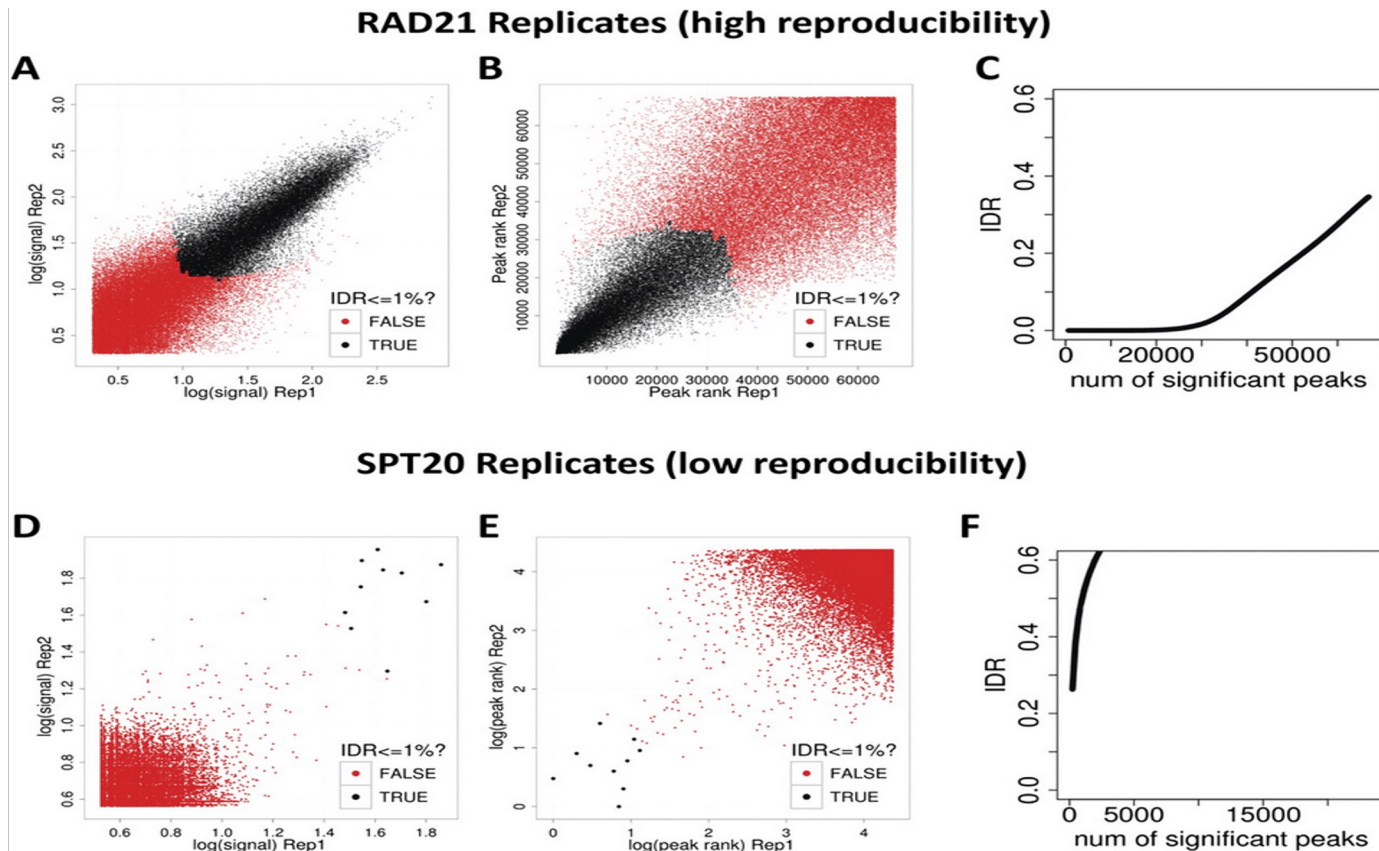
Merge samples prior to the peak calling (e.g recommended by MACS) => “pooling”



IDR - Irreproducible Discovery Rate (ENCODE)

- Measures consistency between replicates
- Uses reproducibility in score rankings between peaks in each replicate to determine an optimal cutoff for significance.
- Idea:
 - The most significant peaks are expected to have high consistency between replicates
 - The peaks with low significance are expected to have low consistency

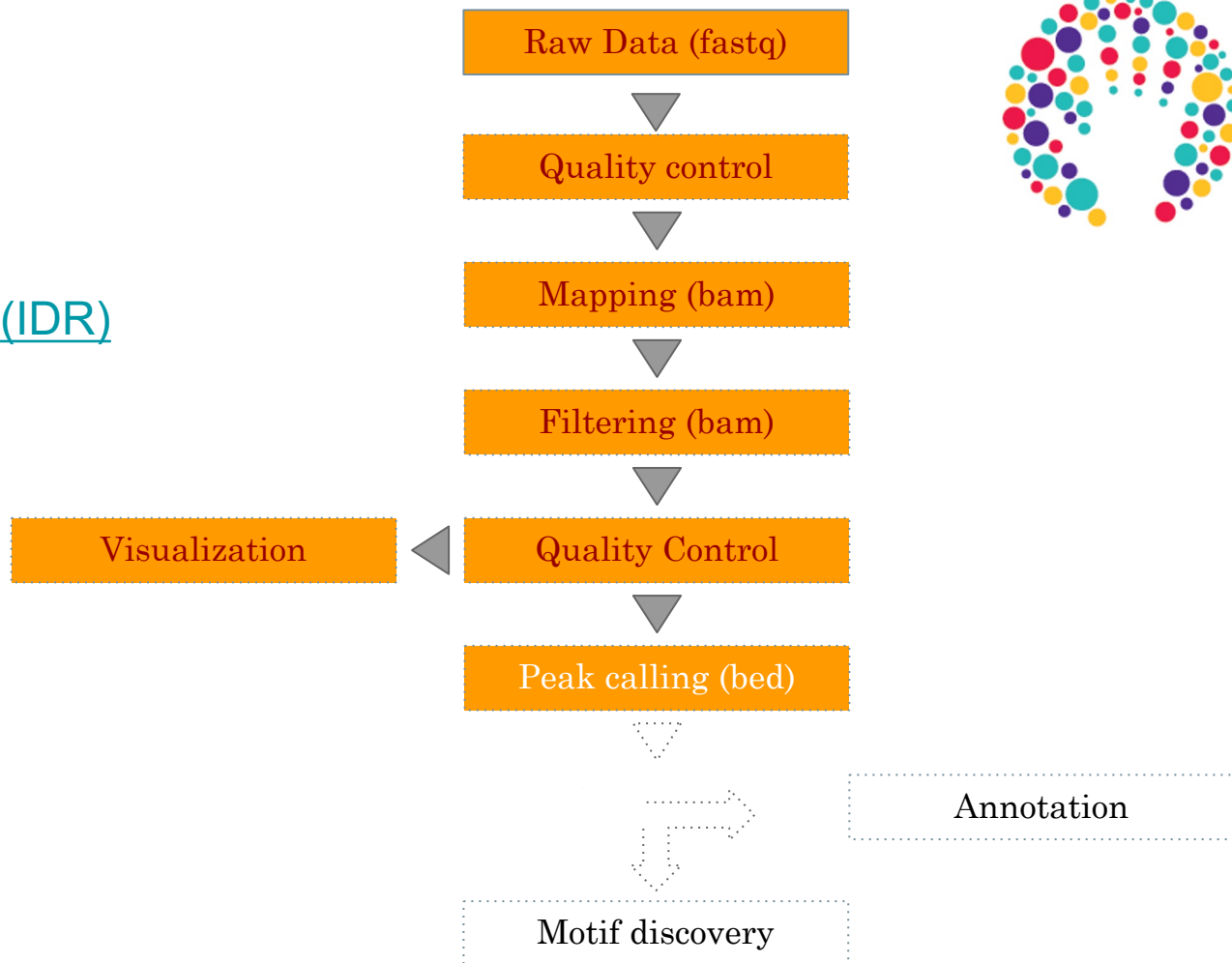
IDR



(!) IDR doesn't work on broad source data!

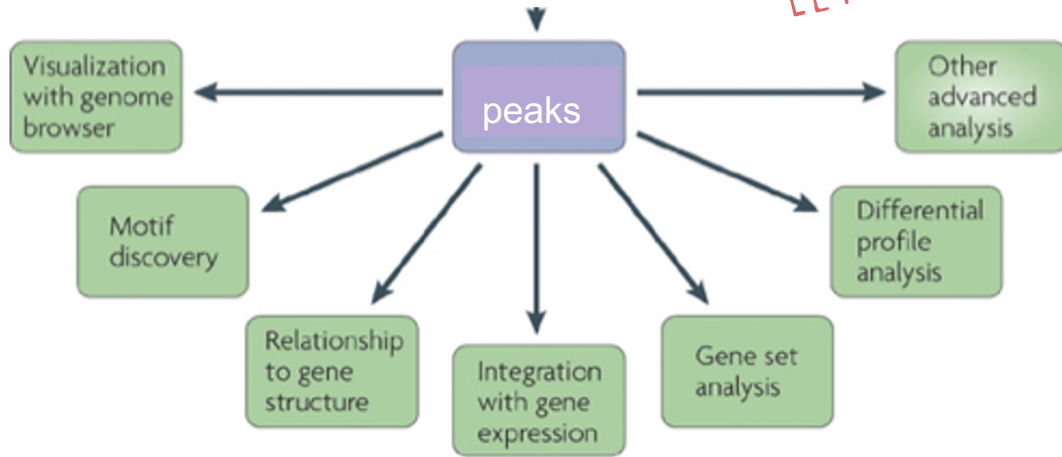
Protocol

- Combine replicate (IDR)

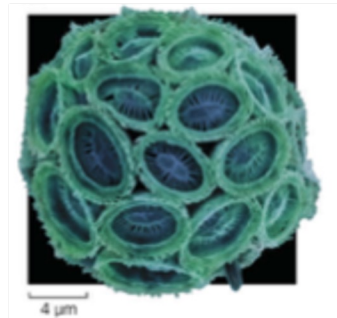
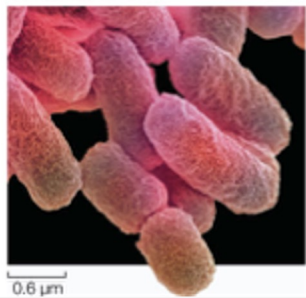


Processing steps are over !

LET'S DO BIOLOGY !!!



Nature Reviews | **Genetics**



What is the biological question ?





What is the biological question ?

« see if you can find something in the data »



What is the biological question ?

~~« see if you can find something in the data »~~

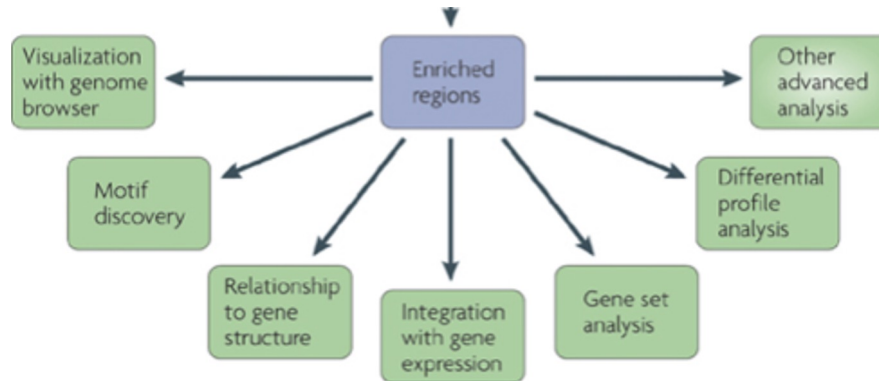
What is the biological question ?

- **Where** do a transcription factor (TF) bind ?
- **How** do a transcription factor (TF) bind ?
 - Which **binding motif(s)** (can be several for a given TF !!)
 - Is the **binding** direct to DNA or via **protein-protein** interactions ?
 - Are there **cofactors** (maybe affecting the motif !!), and if so, identify them
- Which **regulated genes** are directly regulated by a given TF ?
- Where are the **promoters** (PolII) and **chromatin marks** ?

What is the biological question ?

Should drive all « downstream » analyses

Will take time
to « do it all » !!!





What is the biological question ?

What can be the following experimental work ?

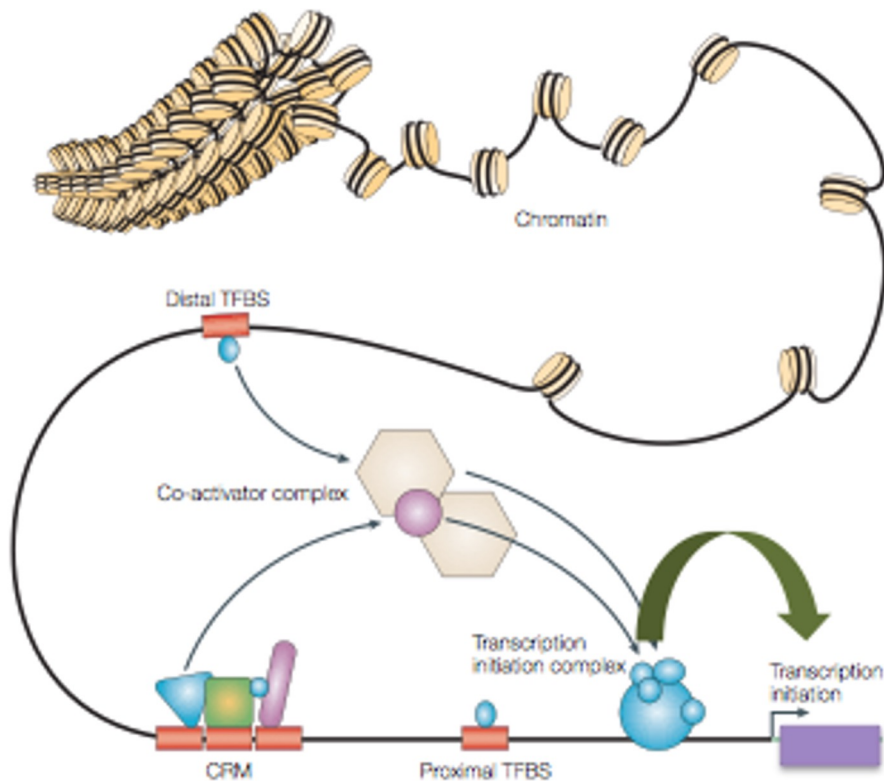
- cell biology (eg: luciferase assay) ?
- in vitro assays (eg: EMSA) ?
- Proteomic (eg: mass spectrometry) ?
- Transgenics ?
- Will depend on
 - the organism
 - available infrastructure



De novo motif discovery

Biological concepts of transcriptional regulation

Transcription factors are proteins that modulate (activate/repress) the expression of **target genes** through the binding on **DNA cis-regulatory elements**

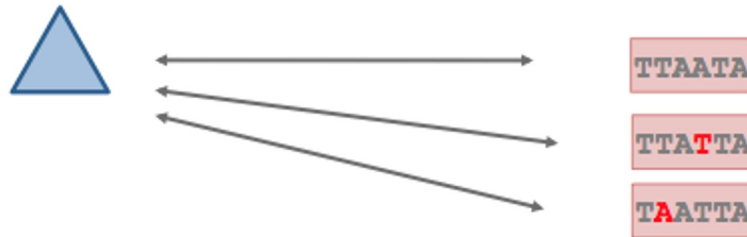


Transcription factor specificity

How do TF « know » where to bind DNA ?



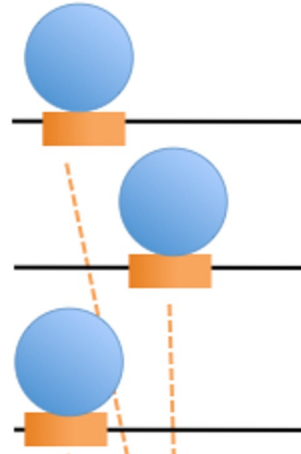
TF recognize TFBS with specific DNA sequences



a given TF is able to bind DNA on TFBSs with different sequences

Binding specificity

transcription factor



cis-regulatory elements



binding motif

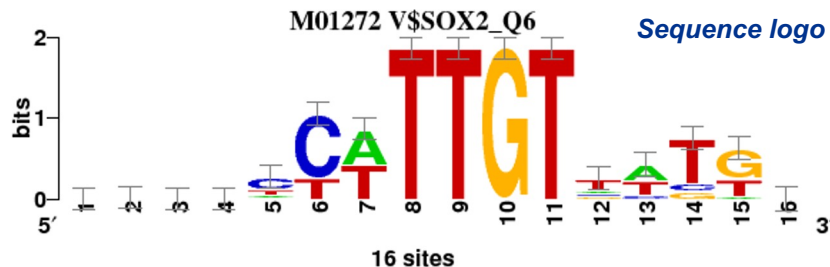
From binding sites to binding motif

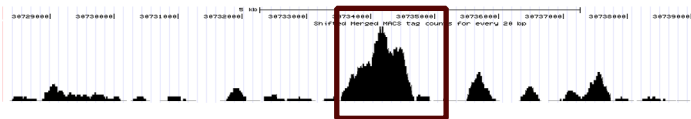
Collection of binding sites used to build the Sox2 matrix (TRANSFAC M01272)

R15133	GCCCTCATTGTTATGC
R15201	AAACTCTTTGTTTGGA
R15231	TTCACCATTGTTCTAG
R15267	GACTCTATTGTCTCTG
R16367	GATATCTTTGTTTCTT
R17099	TGCACCTTTGTTATGC
R19276	AATTCCATTGTTATGA
R19367	AAACTCTTTGTTTGGA
R19510	ATGGACATTGTAATGC
R22342	AGGCCTTTTGTCTCTG
R22344	TGTGCTTTTGTNNNNN
R22359	CTCAACTTTGTAATTT
R22961	GCAGCCATTGTGATGC
R23679	CACCCCTTTGTTATGC
R25928	TTTTCTATTGTTTTTA
R27428	AAAGGCATTGTGTTTC

Position-specific scoring matrix (PSSM)

A	6	7	4	4	2	0	8	0	0	0	0	2	7	0	1	4
C	2	2	6	5	9	12	0	0	0	0	0	2	2	2	0	6
G	4	3	2	4	1	0	0	0	0	16	0	2	0	2	9	3
T	4	4	4	3	4	4	8	16	16	0	16	9	6	11	5	2





ChIP-seq peaks



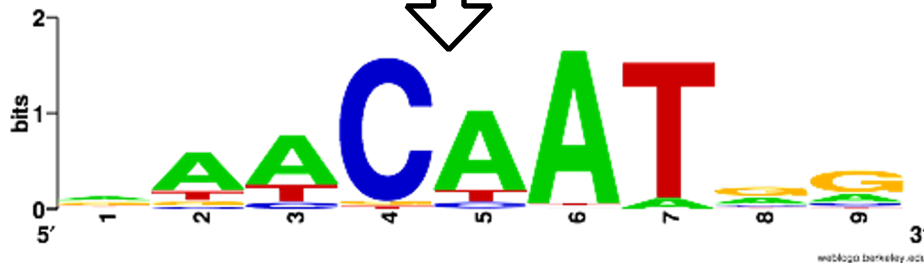
```
>mm9_chr1_39249116_39251316_+
gagaggaaggggagaaagagggagggggagGGTGATAGGTAGCCAGGAG
CCAATGGGGGCGTTTTTCCTTGTCAGGCCACTTGCTGGAATGTGAGATGT
AGAATGACCCAAAGAGAGCTGCCAAGACAGAGCTCTGCCCCAGGAATTGA
ACTCAAAGGGTGTCAGAAAGCAGGTGGCCTTTGTGCACCTGGCGCGGGGA
CGTGGCTCCCTCTTCCGGCTGGTCTAGCCAGGtgccctgcctgcctgcct
gccGTGATCTCTGGACGCCAGTAGAGGGTTGTTGTGGGTTTGGGTGAAAC
ACGCCACCCCTGAGCTCTTCCGCGGGGCTAGCAATCTCCCCATCACCCCA
TTCGCGCTCAGAACCCCTCAGCGAATTAACAGCAGGCCTGGTTCCCCG
```

DNA sequence



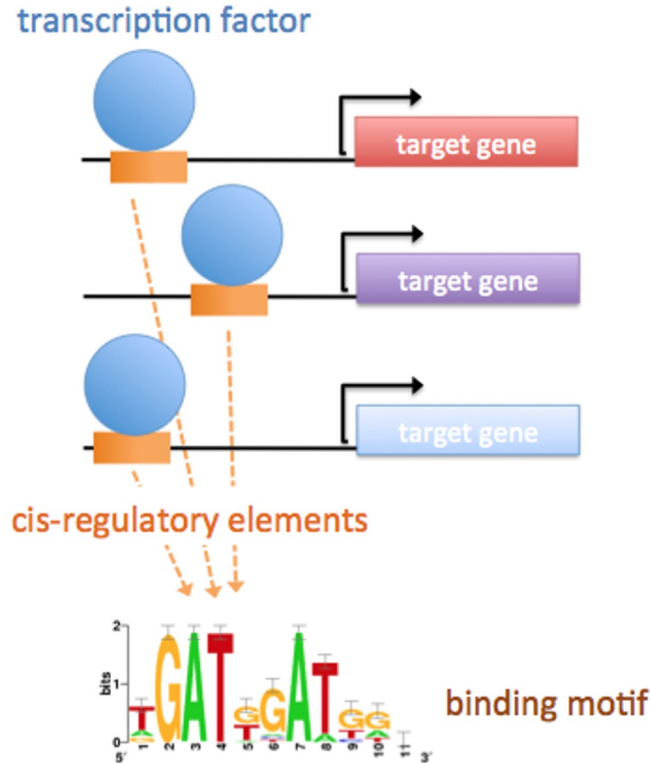
A	[24	54	59	0	65	71	4	24	9]
C	[7	6	4	72	4	2	0	6	9]
G	[31	7	0	2	0	1	1	38	55]
T	[14	9	13	2	7	2	71	8	3]

Discovered motif



Motif logo

De novo motif discovery



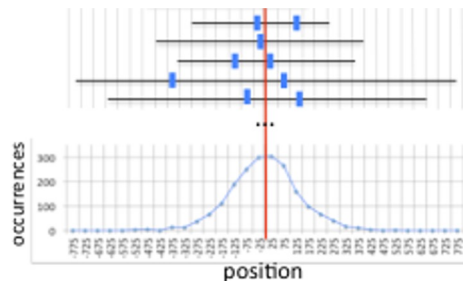
Problem :

*How can we model/describe
the binding specificity of
a given TF ?*

*If there is a common regulating
factor, can we discover its motif
only using these sequences ?*

De novo motif discovery

- Find exceptional motifs based on the sequence only
(No prior knowledge of the motif to look for)
- Criteria of exceptionality:
 - *Over-/under-representation*: higher/lower frequency than expected by chance
 - *Position bias*: concentration at specific positions relative to some reference coordinates (e.g. TSS, peak center, ...).





Some motif discovery tools

- MEME (Bailey et al., 1994)
- RSAT oligo-analysis (van Helden et al., 1998)
- AlignACE (Roth et al. 1998)
- RSAT position-analysis (van Helden et al., 2000)
- Weeder (Pavesi et al. 2001)
- MotifSampler (Thijs et al., 2001)
- ... many others

Why do we need new approaches for genome-wide datasets ?

New approaches for ChIP-seq datasets

- **Size, size, size**
 - limited numbers of promoters and enhancers

↓

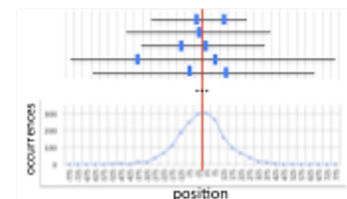
 - dozens of thousands of peaks !!!!!



- **the problem is slightly different**
 - promoters: 200-2000bp from co-regulated genes

↓

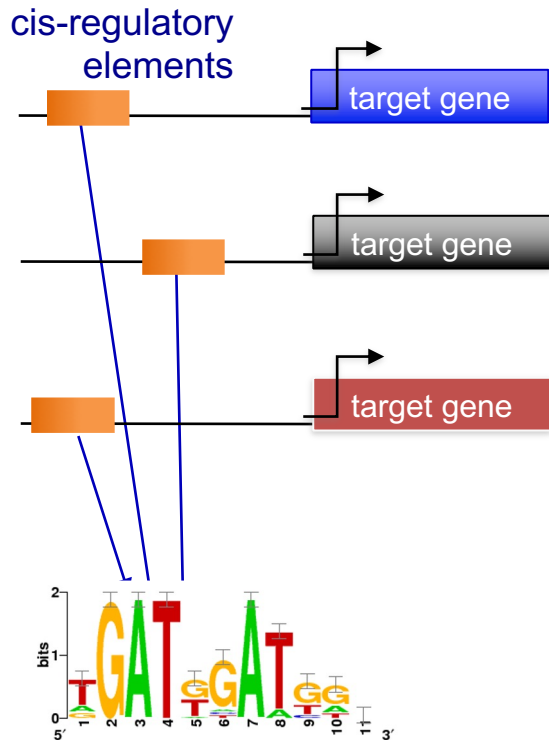
 - peaks: 300bp, positional bias



- **motif analysis: not just for specialists anymore !**
 - complete user-friendly workflows

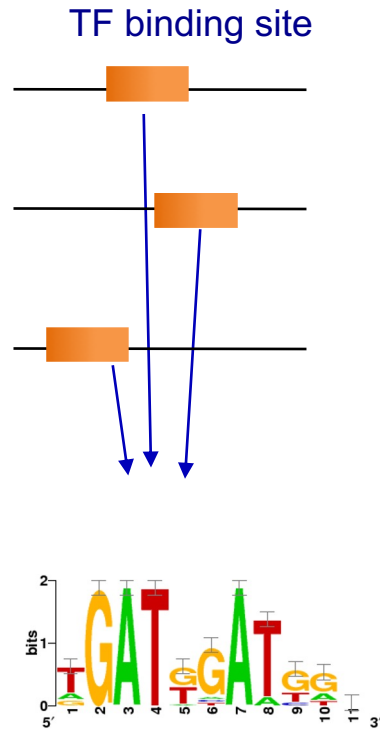
100

Case 1: promoters of co-expressed genes



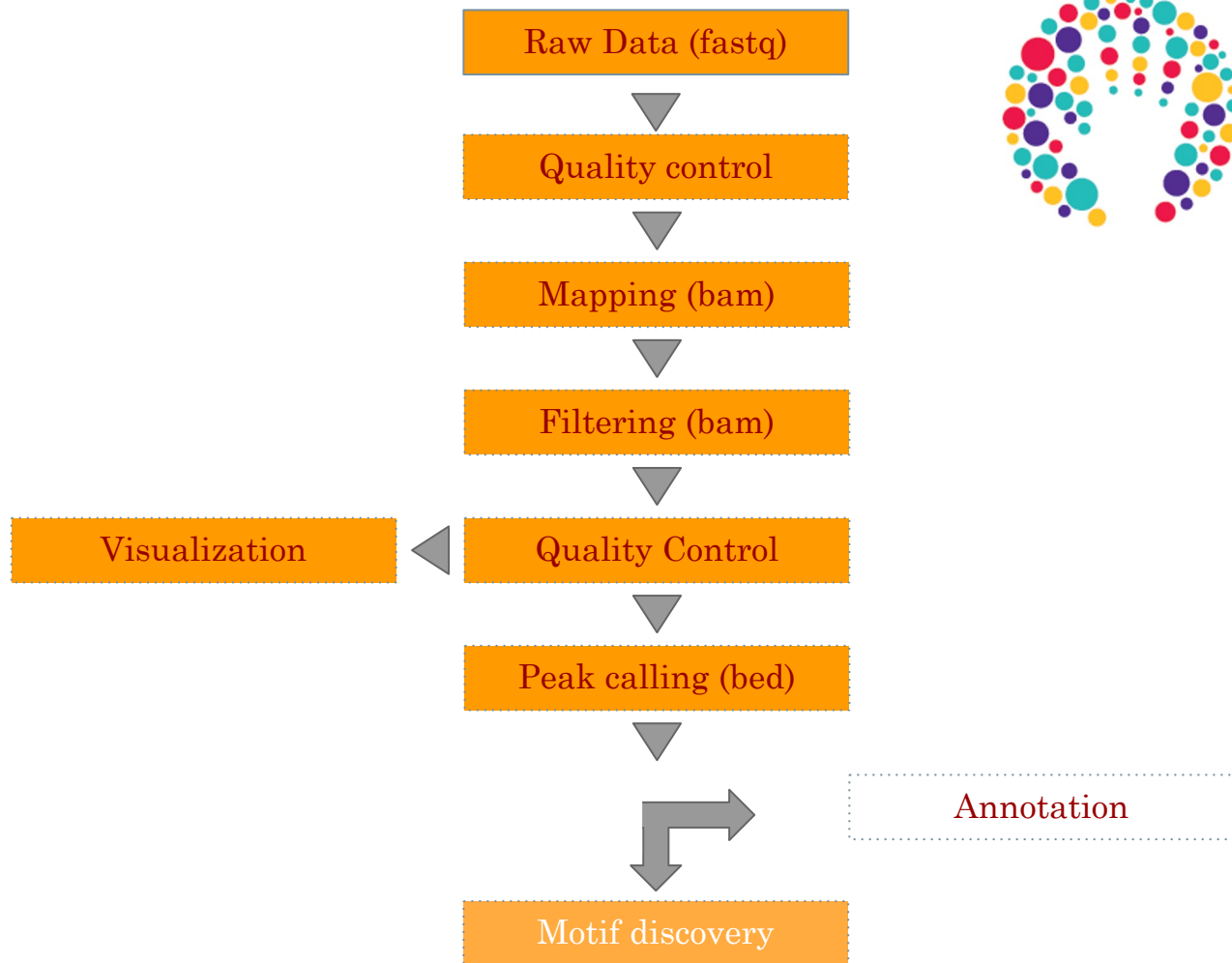
binding motif
(represented as a
sequence logo)

Case 2: ChIP-seq peaks



Protocol

- Motif analysis

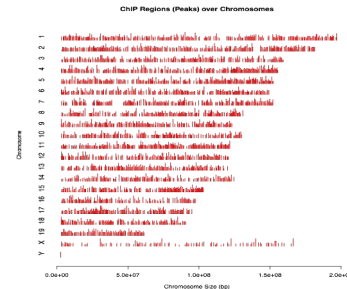
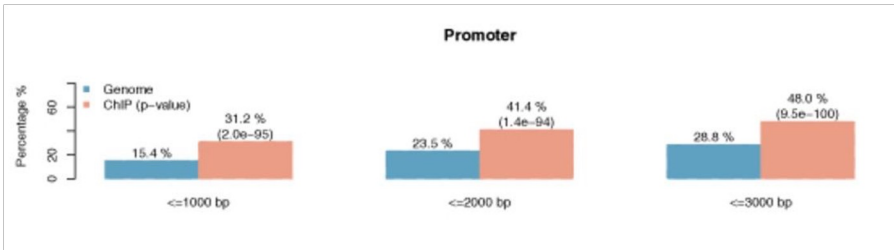




Peak annotation

Are peaks biased towards any genomic features?

- How are the peaks distributed on the chromosomes?
- Are there genomic features (promoters, intergenic, intronic, exonic regions) enriched in the peaks?
- How are the peaks distributed compared to gene structures (TSS, TTS, introns, exons)?
- How are they distributed compared to the genes?



Various tools available

- ChIPseeker (Bioconductor) <https://goo.gl/BemEsw>
- bedtools annotate : <http://bedtools.readthedocs.io>
- HOMER annotatePeaks.pl

Warning : rely on the organism annotation and assembly version

=> not all organisms supported by all programs !

Which are the closest genes?



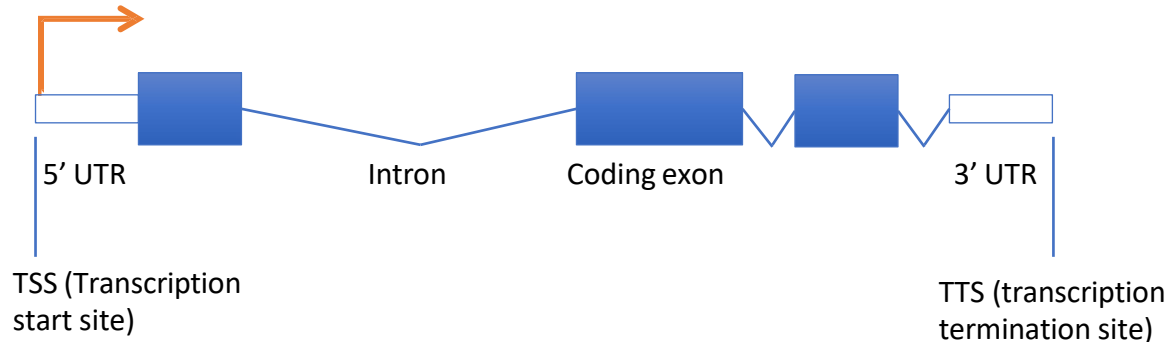
HOMER

Software for motif discovery and ChIP-Seq analysis

HOMER is a well-maintained suite of tools for functional genomics sequencing data sets. It can perform peak-calling and motif analysis, but we will use it for annotation of the peaks only.

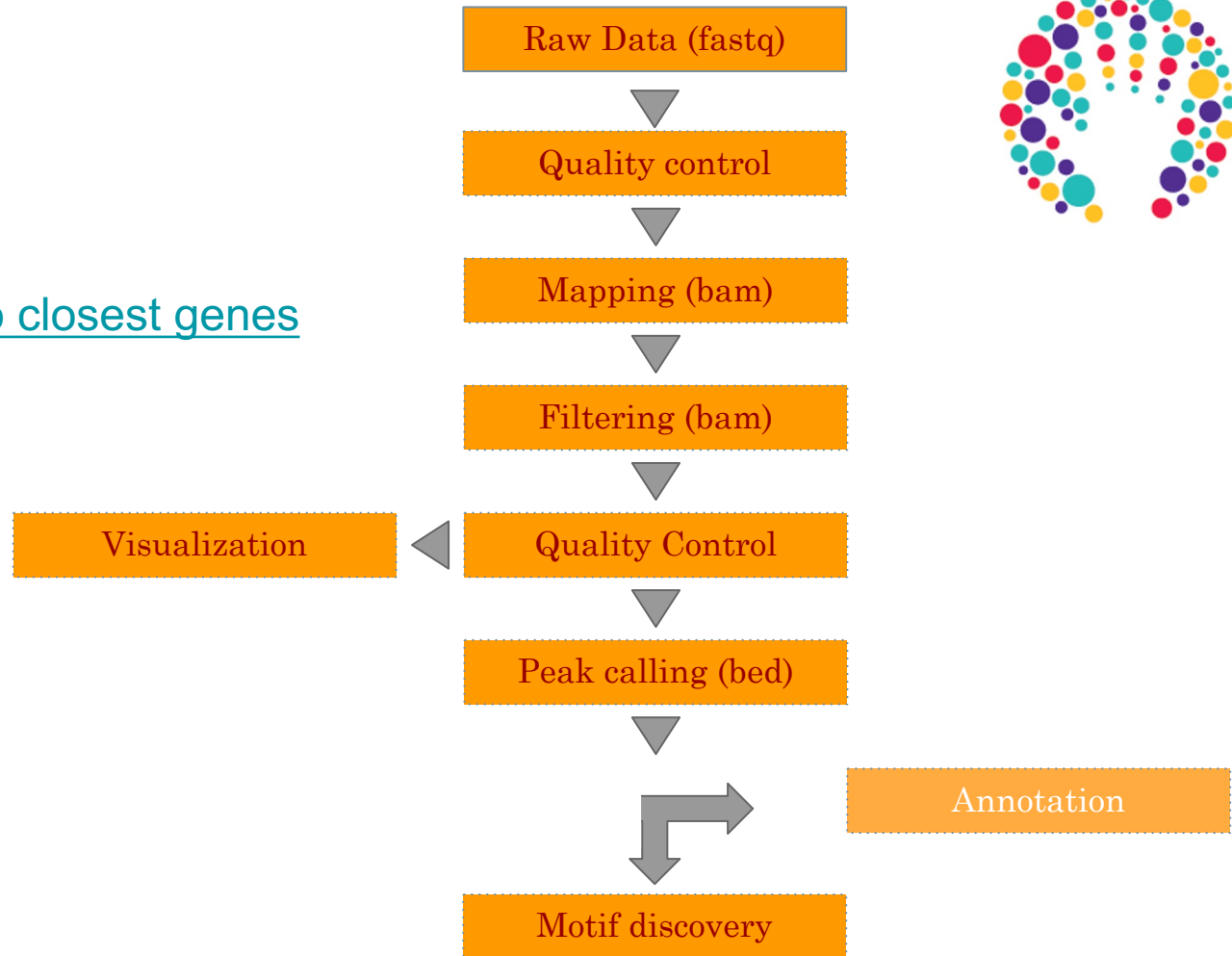
Peak annotation

- Goal: assigning a peak to one or many genome features (genes/transcripts) to understand which genes are possibly regulated by the protein of interest binding
- The name of the gene is important as well as the genic region where the peak is located
- Example of Homer tools:
 - Determines the distance to the nearest Transcription Start Site (TSS) and assigns the peak to that gene
 - Determines the genomic annotation of the region **occupied by the center** of the peak/region.



Protocol

- Associate peaks to closest genes



Now that we have the genes,

Are there some functional categories over-represented ?

ChIP-seq peaks



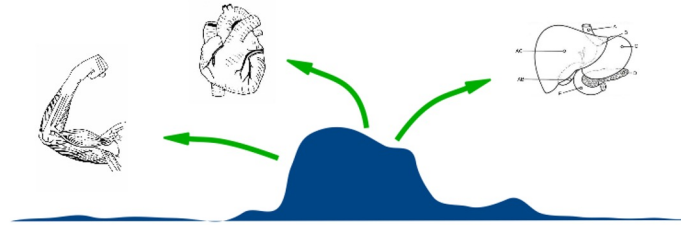
Genes



Ontology terms

GO Molecular Function
GO Biological Process
Disease Ontology
Pathways

...



HOMER [Heinz et al., Mol Cell 2010]

Gene Ontology Analysis of Associated Genes: **annotatePeaks go** option

P-value	LogP	Term	GO Tree	GO ID	# of Genes in Term	# of Target Genes in Term	# of Total Genes	# of Target Genes	Common Genes
2.912e-26	-5.880e+01	immune response	biological process	GO:0006955	349	35	18091	168	Il10,Cd14,Malt1,Ccl2,Ccl7,Ifih:
3.912e-26	-5.850e+01	immune system process	biological process	GO:0002376	679	45	18091	168	S100a9,Egr1,Il10,Cd14,Malt1,C
1.823e-25	-5.696e+01	cytokine activity	molecular function	GO:0005125	178	27	18371	167	Gdf15,Il10,Csf2,Ccl9,Ccl2,Ccl7
3.372e-23	-5.174e+01	defense response	biological process	GO:0006952	430	35	18091	168	Il10,Cd14,Malt1,Nupr1,Ccl2,Cc

Genome Ontology: Looking for Enriched Genomic Annotations: **annotatePeaks genomeOntology** option

Total Input Regions (0.936444051404582.pos): 25961, 33798473 bp

P-value	Log P-value	Annotation	Ann Group	#features	Coverage(bp)	AvgFeatureSize[ref=1301]	Overlap(#peaks)	Overlap(bp)	Expected Overlap(bp, gsize=2.00e+09)	Log Ratio Enrichment	Log P-value(+ underrepresented)	P-value
1e-3660	-8428.50	cpGIsland	basic	28691	21842742	761	9426	5545349	369125	2.71	-8428.50	0.00e+00
1e-2673	-6155.47	promoters	basic	44477	30002652	674	8579	5141062	507021	2.32	-6155.47	0.00e+00
1e-1027	-2363.56	utr5	basic	57703	5423448	93	7002	1516346	91652	2.81	-2363.56	0.00e+00
1e-381	-876.84	exons	basic	503529	73292986	145	9092	3171853	1238595	0.94	-876.84	0.00e+00
1e-341	-783.19	protein-coding	basic	483461	66805131	138	8609	2876255	1128955	0.94	-783.19	0.00e+00
1e-105	-239.83	coding	basic	407555	43508461	106	6592	1482965	735259	0.70	-239.83	6.94e-105
1e-71	-162.42	GC_richLow_complexityLow_complexity	repeats	13724	552081	40	2716	120042	9329	2.55	-162.42	2.89e-71
1e-60	-136.15	miscRNA	basic	11332	4544003	400	592	287477	76790	1.32	-136.15	7.46e-60
3.14e-20	-44.91	tts	basic	44477	28239519	634	1124	718919	477226	0.41	-44.91	3.14e-20
1.21e-10	-22.84	CGnSimple_repeatSimple_repeat	repeats	1241	71601	57	313	15761	1210	2.57	-22.84	1.21e-10
1.06e-09	-20.66	C-richLow_complexityLow_complexity	repeats	9534	1007297	105	673	53335	17022	1.14	-20.66	1.06e-09

GREAT

Species Assembly

- ☐ Human: GRCh37 ([UCSC hg19, Feb/2009](#))
☐ Mouse: NCBI build 37 ([UCSC mm9, Jul/2007](#))
☐ Mouse: NCBI build 38 ([UCSC mm10, Dec/2011](#))
☐ Zebrafish: Wellcome Trust Zv9 ([danRer7, Jul/2010](#)) [Zebrafish CNE set](#)

Can I use a different species or assembly?

Test regions

- ☒ BED file:
☐ BED data:

What should my test regions file contain?

How can I create a test set from a UCSC Genome Browser annotation track?

Background regions

- ☒ Whole genome
☐ BED file:
☐ BED data:

When should I use a background set?

What should my background regions file contain?

Association rule settings

Show settings »

Submit

Reset

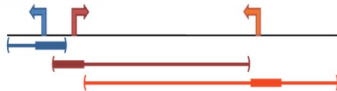
Note: Only human (hg19), mouse (mm9, mm10) and zebrafish (danRer7) genomes are supported

GREAT

Associating genomic regions with genes

GREAT calculates statistics by associating genomic regions with nearby genes and applying the gene annotations to the regions. Association is a two step process. First, every gene is assigned a regulatory domain. Then, each genomic region is associated with all genes whose regulatory domain it overlaps.

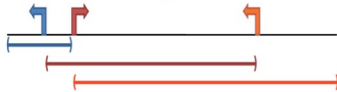
☒ Basal plus extension



Proximal: kb upstream, kb downstream, plus Distal: up to kb

Gene regulatory domain definition: Each gene is assigned a basal regulatory domain of a minimum distance upstream and downstream of the TSS (regardless of other nearby genes). The gene regulatory domain is extended in both directions to the nearest gene's basal domain but no more than the maximum extension in one direction.

☐ Two nearest genes



within kb

Gene regulatory domain definition: Each gene is assigned a regulatory domain that extends in both directions to the nearest gene's TSS but no more than the maximum extension in one direction.

☐ Single nearest gene



within kb

Gene regulatory domain definition: Each gene is assigned a regulatory domain that extends in both directions to the midpoint between the gene's TSS and the nearest gene's TSS but no more than the maximum extension in one direction.

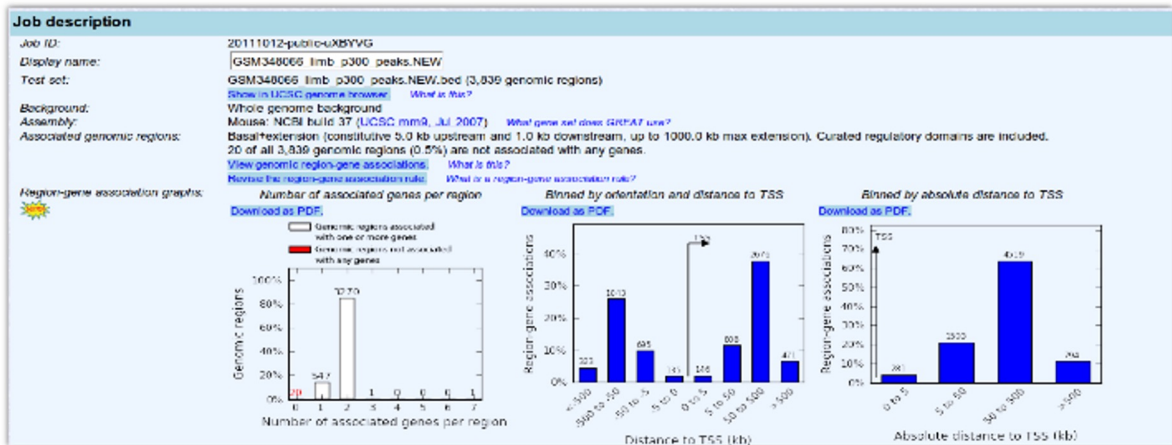


Gene Transcription Start Site (TSS)

Note: Only human (hg19), mouse (mm9, mm10) and zebrafish (danRer7) genomes are supported

GREAT

- Input
 - bed file with peaks
- Output
 - Enriched GO terms and functions



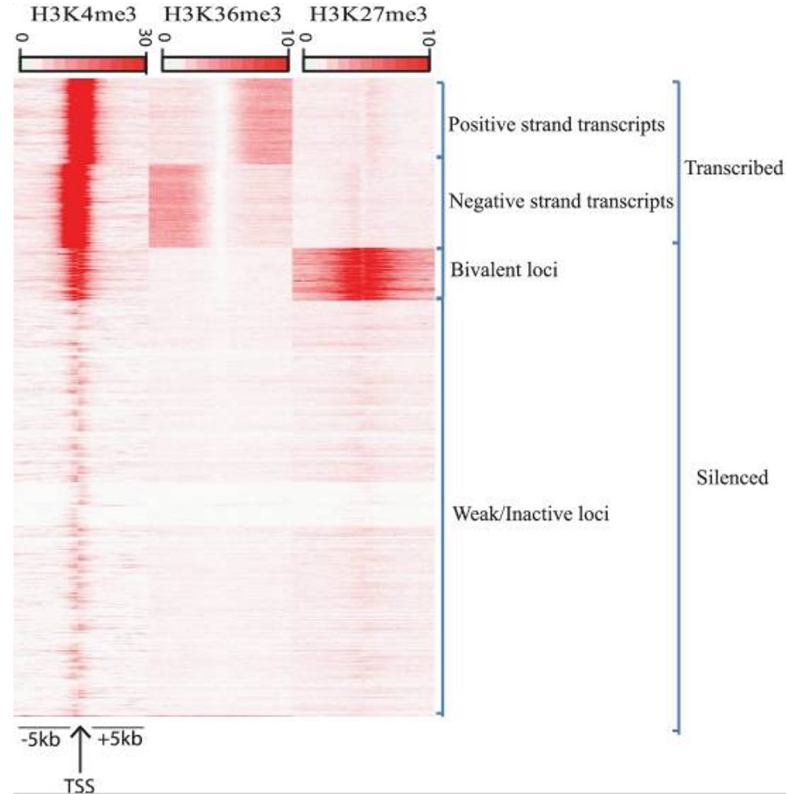
Mouse Phenotype

Table controls: Export Shown top rows in this table: 20 Term annotation count: Min: 1 Max: Inf Set

Term Name	Binom Rank	Binom Raw P-Value	Binom FDR Q-Val	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	Hyper FDR Q-Val	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
abnormal limbs/digits/tail morphology	2	2.0559e-91	6.6837e-88	2.1465	780	20.32%	6	2.5295e-40	2.2020	278	681	8.31%
abnormal craniofacial morphology	3	9.3822e-91	2.0334e-87	2.0082	897	23.10%	10	8.9231e-36	2.0382	297	766	8.86%
abnormal limb morphology	5	2.4990e-80	3.2497e-77	2.3077	634	15.73%	9	7.4787e-37	2.4541	202	444	6.04%
abnormal appendicular skeleton morphology	10	3.0255e-70	1.9672e-67	2.3450	517	13.47%	17	3.9549e-30	2.4098	172	385	5.14%
abnormal skeleton extremities morphology	12	3.2687e-69	1.7711e-66	2.3724	499	13.00%	21	7.0557e-29	2.4222	163	363	4.87%
abnormal paw/hand/foot morphology	13	4.0300e-69	2.0156e-66	2.6813	404	10.52%	23	5.4918e-28	2.7186	126	250	3.77%
abnormal head morphology	14	6.4657e-67	3.0029e-64	2.0134	672	17.50%	25	2.9042e-27	2.0982	223	585	6.67%
abnormal digit morphology	18	1.0543e-61	3.8084e-59	2.6982	358	9.33%	36	1.2033e-25	2.7998	109	210	3.26%
abnormal cartilage morphology	23	7.3728e-58	2.0843e-55	2.3432	430	11.20%	29	1.1337e-26	2.5089	140	301	4.19%
abnormal skeleton development	24	3.5769e-56	9.6904e-54	2.0833	530	13.81%	38	5.2377e-25	2.1414	185	466	5.53%
abnormal long bone morphology	25	4.6593e-56	1.2118e-53	2.3374	419	10.91%	43	4.9983e-24	2.3823	140	317	4.19%

Other analyses

- Clustering peaks
(DeepTools, HOMER, seqMINER)



Ye et al, 2011

The darker the red the higher the read enrichment

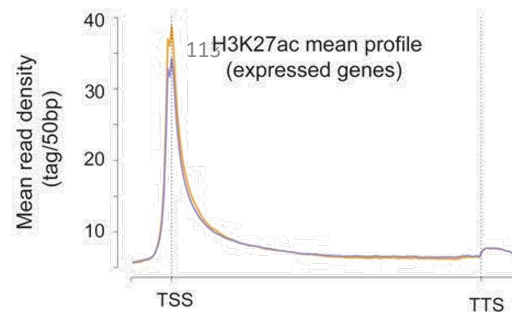
Meta-profiles / Clustering`

- Global visualization of the data
- Need:
 - Regions of interest
 - Regions around a reference point e.g all TSS +/- 1Kb,...
 - Scaled regions e.g peaks, gene bodies,...
 - Signal data (mapped reads)
- Data can be grouped together with clustering methods such as k-means.

Heatmap

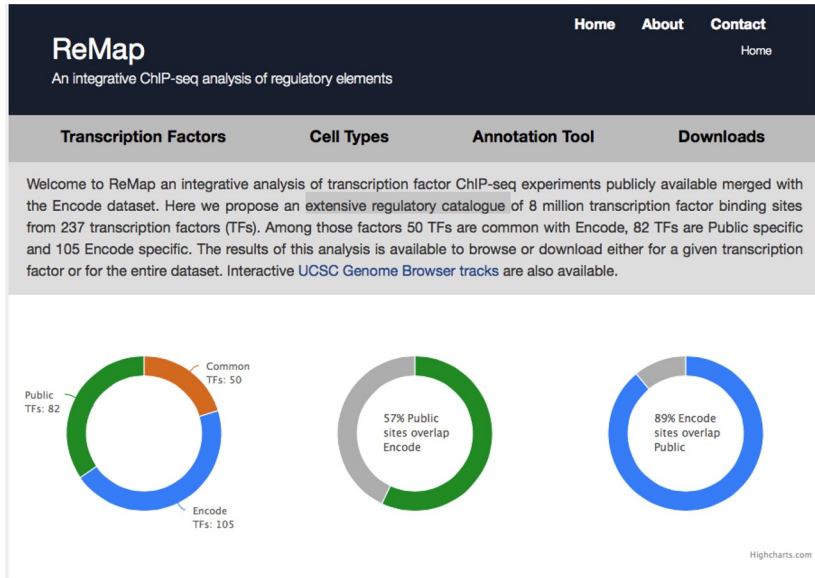


Mean profile

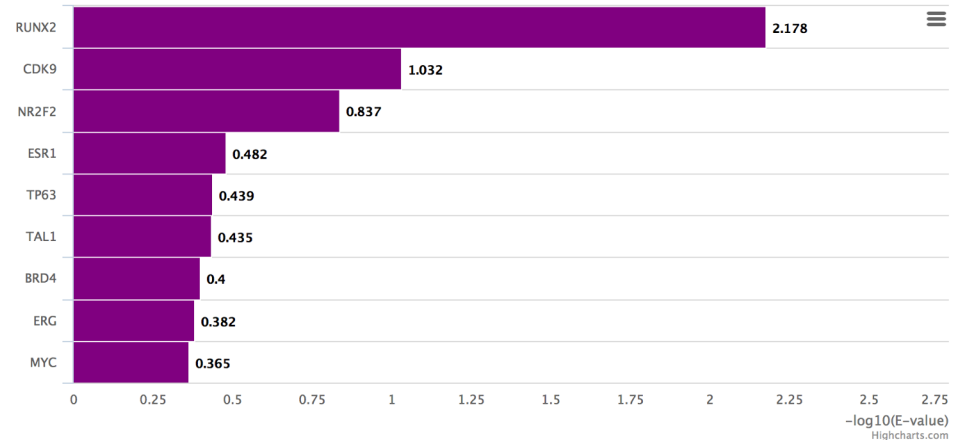


Other analyses

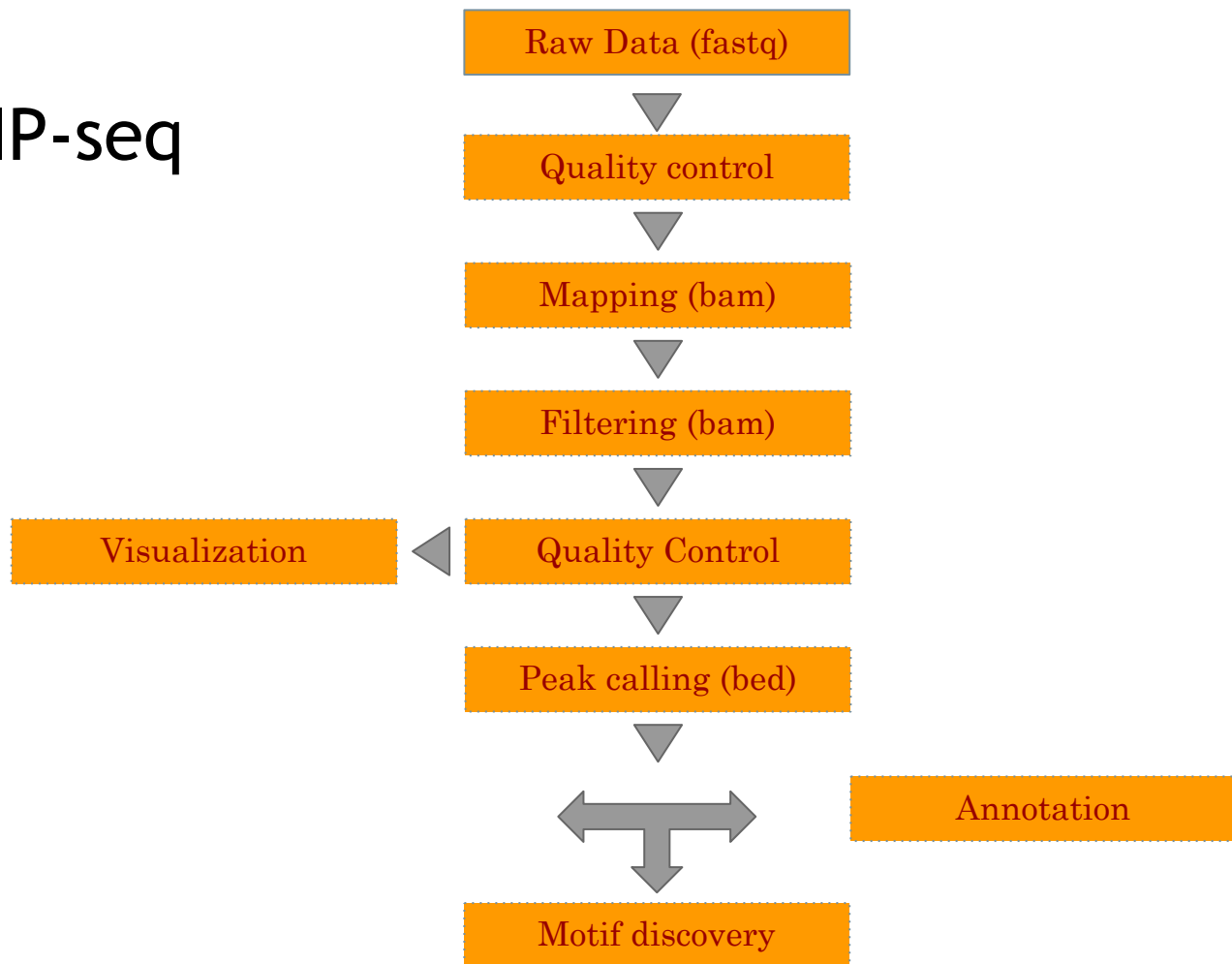
- ReMAP (<http://tagc.univ-mrs.fr/remap/>)
 - Is my peak dataset enriched for known TF peaks?



Enriched TFs in intersection



Overview of ChIP-seq Pipeline



Identify open chromatin regions

Chromatin accessibility assays

- Chromatin accessibility is the degree to which nuclear macromolecules are able to physically contact chromatinized DNA and is determined by the occupancy and topological organization of nucleosomes as well as other chromatin-binding factors that occlude access to DNA (Klemm et al, 2019)
- Open chromatin regions contains generally transcriptionally active genes
- The accessible genome comprises ~2–3% of total DNA sequence yet captures more than 90% of regions bound by TFs (Thurman et al, 2012)
- Chromatin accessibility is measured by quantifying the susceptibility of chromatin to either enzymatic methylation or cleavage of its constituent DNA
- Chromatin accessibility assays (non exhaustive list)
 - FAIRE-seq
 - DNase-seq
 - MNase-seq
 - ATAC-seq

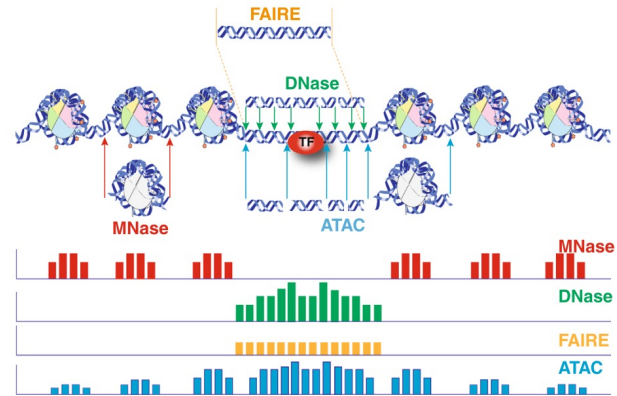
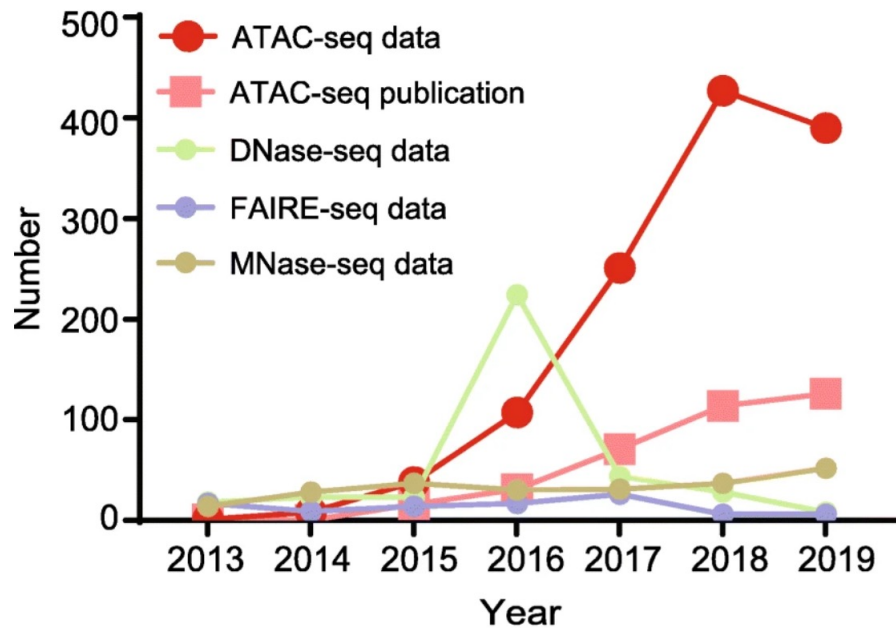


Figure 1 Schematic diagram of current chromatin accessibility assays performed with typical experimental conditions. Representative DNA fragments generated by each assay are shown, with end locations within chromatin defined by colored arrows. Bar diagrams represent data signal obtained from each assay across the entire region. The footprint created by a transcription factor (TF) is shown for ATAC-seq and DNase-seq experiments.

(Tsompana and Buck, 2014)

Chromatin accessibility assays

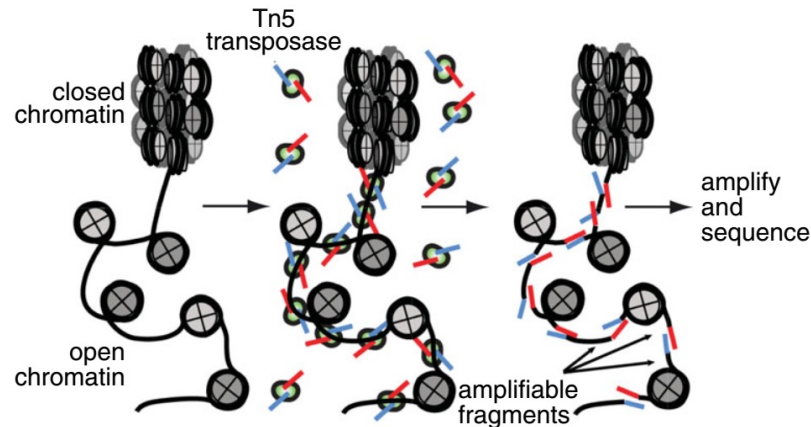
- ATAC-seq has become the most widely used method to detect and analyze open chromatin regions



Yan et al, 2020

ATAC-seq

- Buenrostro et al, 2013
- ATAC-seq is a method for determining chromatin accessibility across the genome
- Transcription factor binding sites and positions of nucleosomes can be identified from the analysis of ATAC-Seq
- Advantages of ATAC-seq over other chromatin accessibility assays:
 - The sensitivity and specificity are comparable to DNase-seq but superior to FAIRE-seq
 - Straightforward and rapidly implemented protocol. ATAC-seq libraries can be generated in less than 3 hours
 - Low number of cells required (500 - 50,000 cells)
 - single-cell ATAC-seq (scATAC-seq) protocol (Cusanovich et al, 2015)



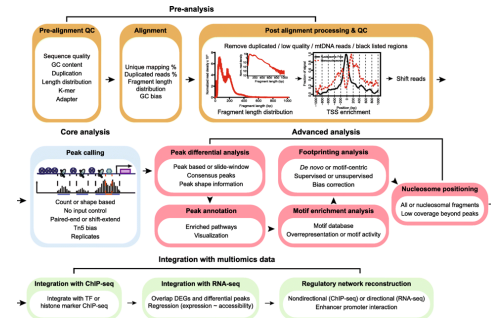
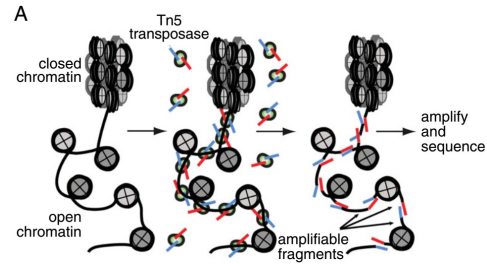
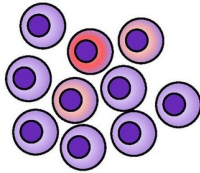
(Buenrostro et al., 2015).

ATAC-seq process

Sample processing

ATAC-seq

Data analysis

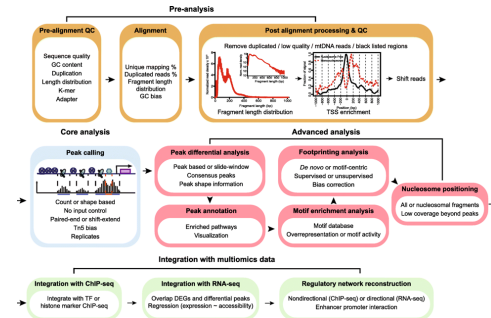
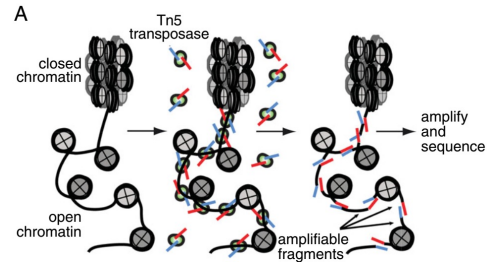
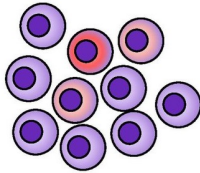


ATAC-seq

Sample processing

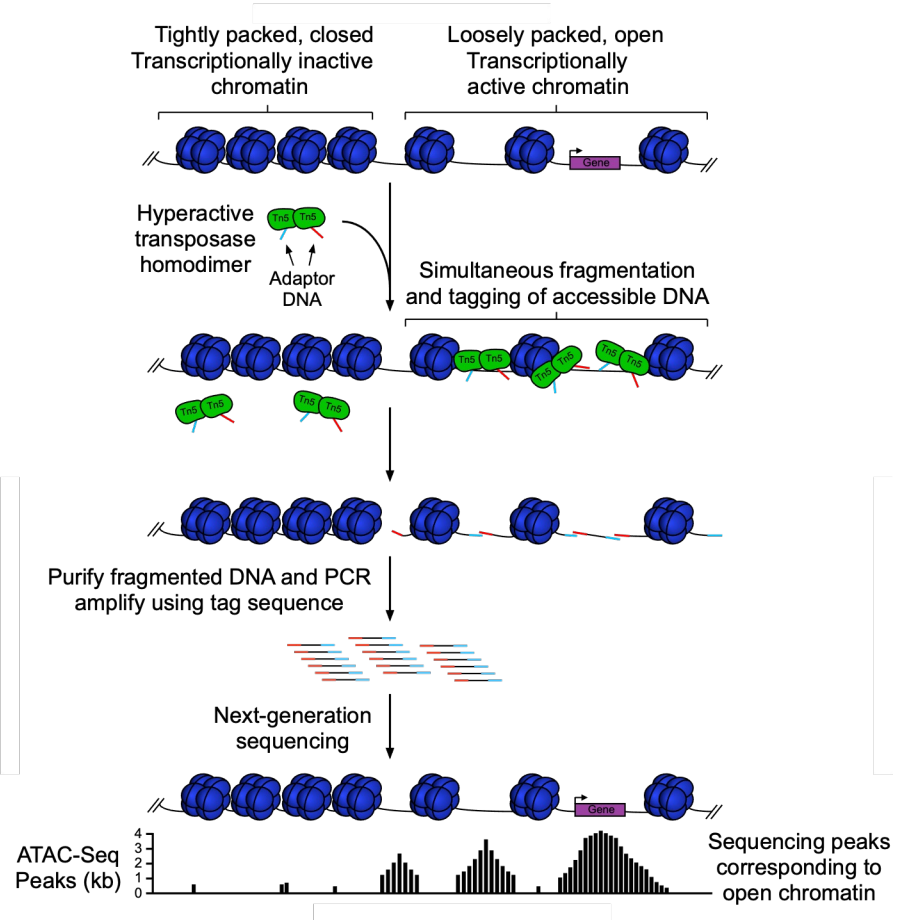
ATAC-seq

Data analysis



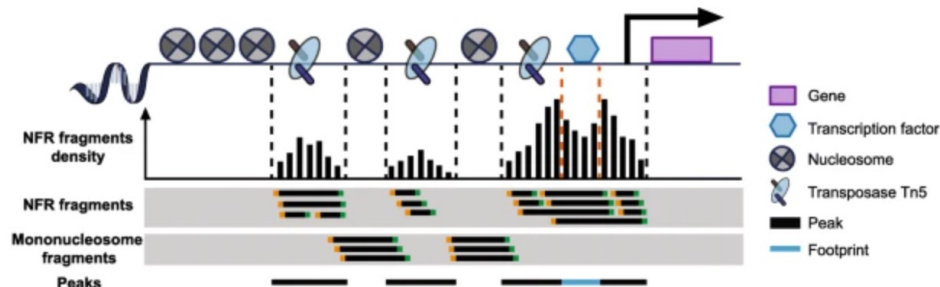
ATAC-seq

- ATAC-seq protocol utilizes a hyperactive Tn5 transposase to insert sequencing adapters into open chromatin regions
- In a process called "tagmentation", Tn5 transposase cleaves and tags double-stranded DNA with sequencing adaptors
- No additional library prep is needed
- Expected results are enrichments of sequenced reads in open chromatin regions as closed chromatin regions, DNA regions bound by TFs or wrapped around nucleosomes should be protected from Tn5 cleavage activity.

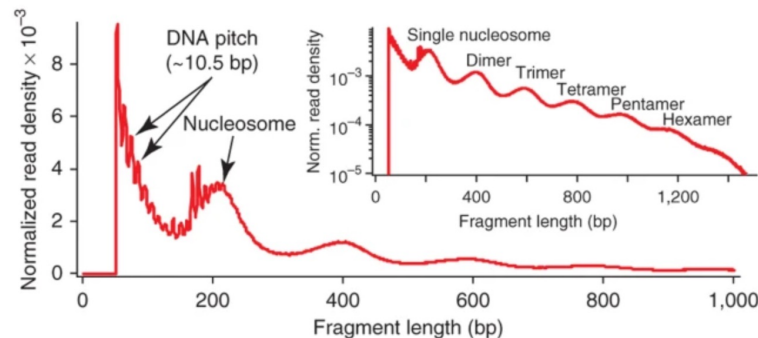


ATAC-seq

- **Paired-end sequencing** so that by looking at the distance between the two reads of a pair, we know in which the chromatin environment (Nucleosome Free Region (NFR), around a mono, di-, nucleosome, around a TF) of the DNA fragment.



Yan et al, 2020



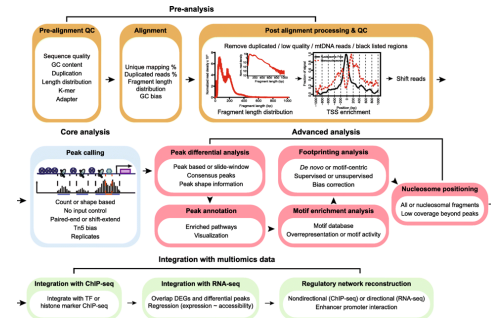
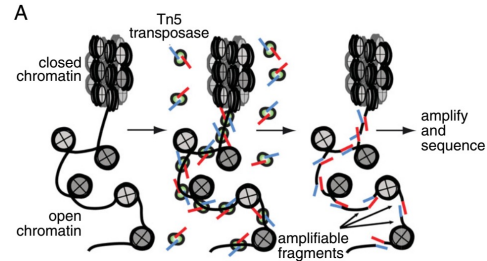
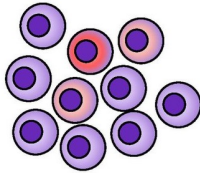
Buenrostro et al, 2023^{1,2,3}

Analysis of ATAC-seq data

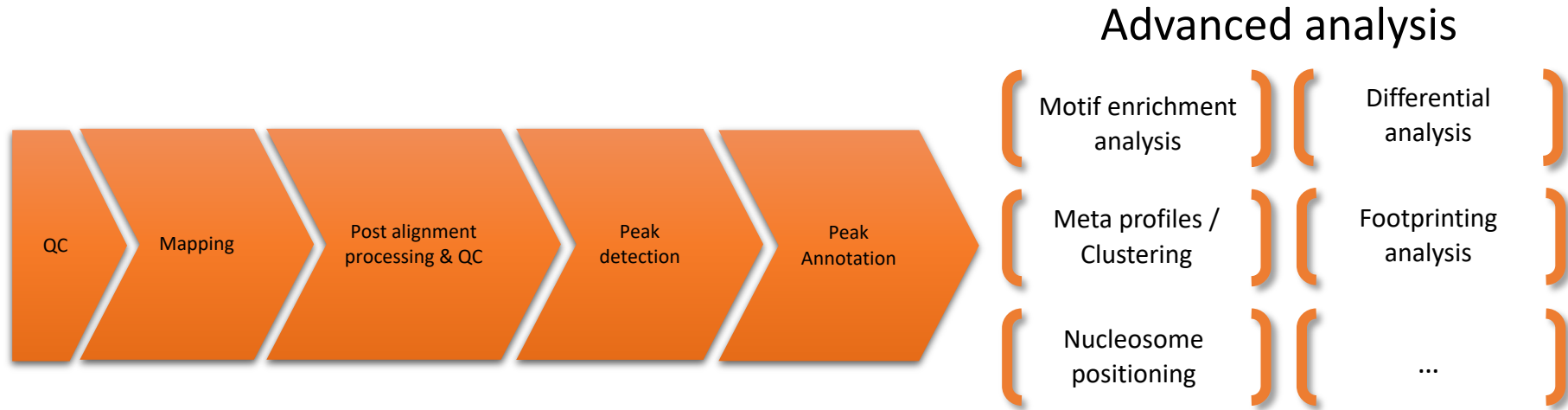
Sample processing

ATAC-seq

Data analysis

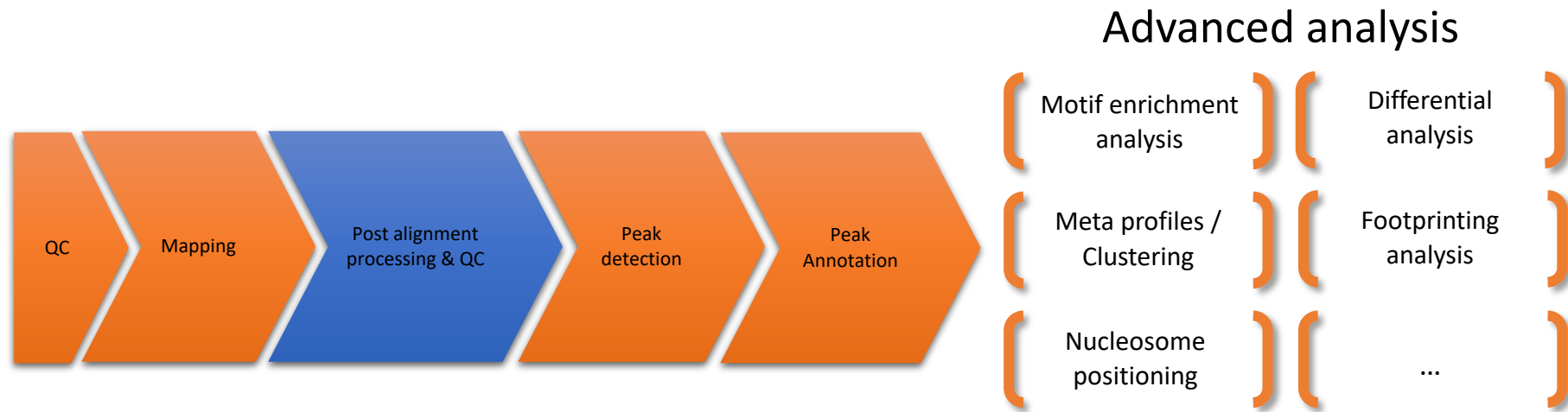


Analysis of ATAC-seq data



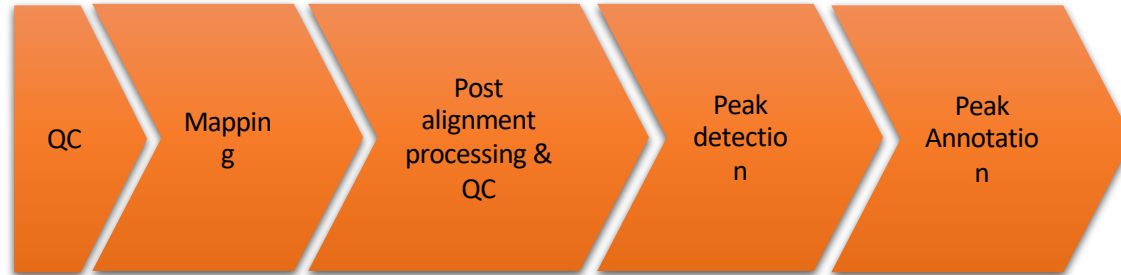
- Overall analysis resemble ChIP-seq data analysis
- Description of particularities of ATAC-seq data analysis

Analysis of ATAC-seq data

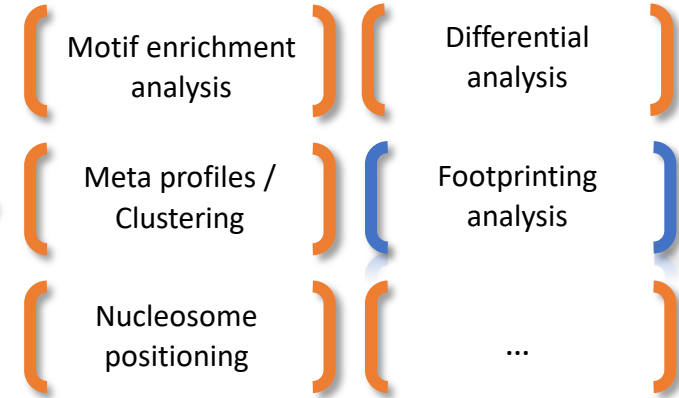


- Some cleaning steps are required for ATAC-seq. For example:
 - A large percentage of reads are derived from mitochondrial DNA. These reads are removed as mitochondrial genome is generally not of interest.
 - Omni-ATAC (Corces et al, 2017)

Analysis of ATAC-seq data

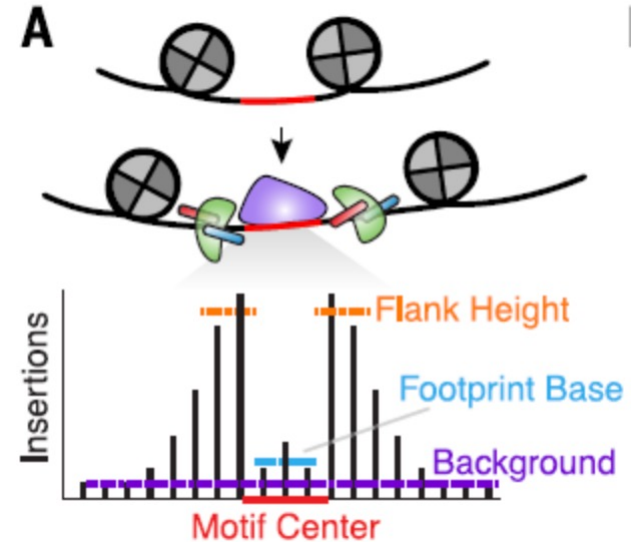


Advanced analysis



Footprinting analysis

- Tn5 cuts in open chromatin regions
- DNA is protected from cleavage at position of TF binding creating a “notch” in ATAC-seq signal
- Footprinting analysis identifies TF activities
 - Height of the notch reflects TF activity
 - Compare TF activity between different conditions

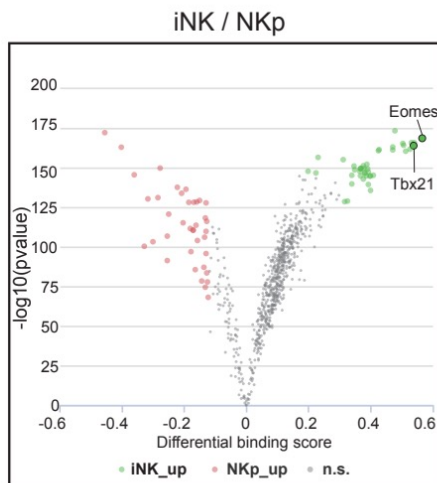


Corces et al, 2019

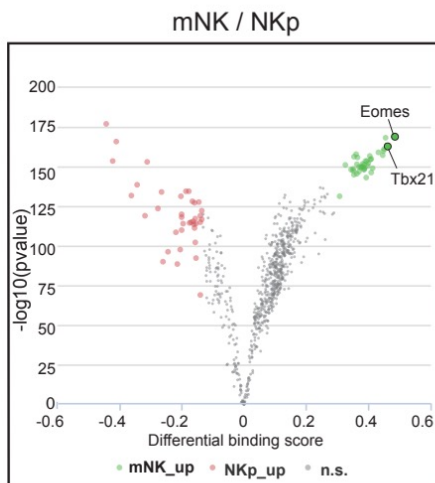
Footprinting analysis

- Volcano plots showing differential TF binding activity as predicted by TOBIAS footprinting analysis in ATAC-seq data of NKp, iNK and mNK from Shin et al. (c) iNK vs NKp; (d) mNK vs NKp; (e) mNK vs iNK.
- Each dot represents a TF
- TFs which activity is changing between the two compared developmental stages are colored (see color legend below volcano plots)

c



d



e

